The allegory axioms, and what defines them in the Frobenius calculus

§1. Notation

Traditionaly Adjunction F -| G is defined as $F(x) \le y$ iff $x \le G(y)$, and you derive laws like this:

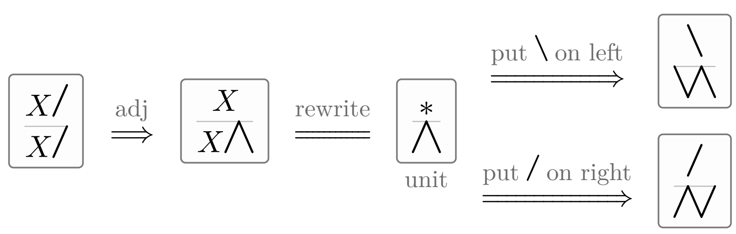
$$\begin{array}{ccccc} F(x) \le F(x) & \implies & x \le G(F(x)) & \implies & F(x) \le F(G(F(x))) \\ G(y) \le G(y) & \implies & F(G(y)) \le y & \implies & G(F(G(y))) \le G(y) \end{array}$$

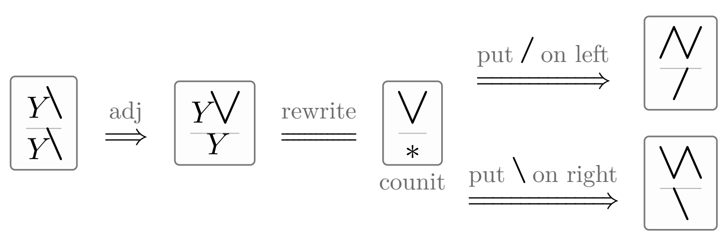
By define $X : * \mapsto x$, $Y : * \mapsto y$ and use diagram order, application can be replaced by composition: $XF \le Y$ iff $X \le YG$

(1a)

- write F as $/$, G as $\backslash$
- render $a \le b$ as $\boxed{\begin{array}{c} a \\ b \end{array}}$
- **rewrite** $X \le XF$ as $1 \le F$

$$\boxed{\begin{array}{c} X/ \\ \hline Y \end{array}} = \boxed{\begin{array}{c} X \\ \hline Y\backslash \end{array}} \qquad \text{adj}$$





now we have $/\backslash/ = /$ and $\backslash/\backslash = \backslash$, and you just discovered string diagrams.

§1.1. Adjunctions

(1.1a)

F ⊣ G	F’s type	monad FG	𝟙⊑FG	GF⊑𝟙	F preserves ∪	G preserves ∩	FGF=F	GFG=G
◁ ⊣ ▷	A → A⊗A	◁▷=𝟙	𝟙⊑◁▷	▷◁⊑𝟙	(R∪S)◁= R◁∪S◁	(R∩S)▷= R▷∩S▷	◁▷◁=◁	▷◁▷=▷
⊸ ⊣ ⟜	A → 𝕀	⊸⟜=⊤	𝟙⊑⊸⟜	⟜⊸⊑𝟙	(R∪S)⊸= R⊸∪S⊸	(R∩S)⟜= R⟜∩S⟜	⊸⟜⊸=⊸	⟜⊸⟜=⟜
° ⊣ °	(A⇸B) → (B⇸A)	°°=𝟙	R=R°°	R=R°°	(R∪S)°= R°∪S°	(R∩S)°= R°∩S°	R°°°=R°	R°°°=R°
⟜◁ ⊣ ▷⊸	(X⊗A → Y) → (X → Y⊗A)	𝟙	(⟜◁⊗𝟙) (𝟙⊗▷⊸)=𝟙	(𝟙⊗⟜◁) (▷⊸⊗𝟙)=𝟙	=	=	=	=
▷⊸ ⊣ ⟜◁	(X → Y⊗A) → (X⊗A → Y)	𝟙	(𝟙⊗⟜◁) (▷⊸⊗𝟙)=𝟙	(⟜◁⊗𝟙) (𝟙⊗▷⊸)=𝟙	=	=	=	=
Δ ⊣ ∩	(A⇸B) → (A⇸B)^2	R↦R∩R=R	R⊑R∩R	R∩S⊑R	Δ(R∪S)= ΔR∪ΔS	(R∩T)∩(S∩U)= (R∩S)∩(T∩U)	R∩R=R	R∩R=R
∪ ⊣ Δ	(A⇸B)^2 → (A⇸B)	(R,S)↦ (R∪S,R∪S)	R⊑R∪S	R∪R⊑R	(R∪T)∪(S∪U)= (R∪S)∪(T∪U)	Δ(R∩S)= ΔR∩ΔS	R∪R=R	R∪R=R
⊥ ⊣ !	{*} → (A⇸B)	—	—	⊥⊑R	—	—	—	—
·S ⊣ /S	(A⇸B) → (A⇸C)	S/S	R⊑(RS)/S	(T/S)S⊑T	(R∪T)S= RS∪TS	(R∩T)/S= R/S∩T/S	((RS)/S)S =RS	((T/S)S)/S =T/S
S· ⊣ S\	(B⇸C) → (A⇸C)	S\S	R⊑S\(SR)	S(S\T)⊑T	S(R∪T)= SR∪ST	S\(R∩T)= S\R∩S\T	S(S\(SR)) =SR	S\(S(S\T)) =S\T
R∩ ⊣ R⇒	(A⇸B) → (A⇸B)	X↦R⇒(X∩R)	X⊑R⇒(X∩R)	R∩(R⇒Y)⊑Y	R∩(X∪Y)= (R∩X)∪(R∩Y)	R⇒(X∩Y)= (R⇒X)∩(R⇒Y)	R∩(R⇒(X∩R)) =X∩R	R⇒(R∩(R⇒Y)) =R⇒Y
𝓓 ⊣ ·⊤	(A⇸B) → Cor A	R↦(𝓓R)⊤	R⊑(𝓓R)⊤	𝓓(A⊤)⊑A	𝓓(R∪S)= 𝓓R∪𝓓S	(A∩B)⊤= A⊤∩B⊤	𝓓((𝓓R)⊤) =𝓓R	(𝓓(A⊤))⊤ =A⊤
𝓡 ⊣ ⊤·	(A⇸B) → Cor B	R↦⊤(𝓡R)	R⊑⊤(𝓡R)	𝓡(⊤A)⊑A	𝓡(R∪S)= 𝓡R∪𝓡S	⊤(A∩B)= ⊤A∩⊤B	𝓡(⊤(𝓡R)) =𝓡R	⊤(𝓡(⊤A)) =⊤A
·f ⊣ ·f°	(A⇸B) → (A⇸C)	ff°	𝟙⊑ff°	f°f⊑𝟙	(R∪S)f= Rf∪Sf	(R∩S)f°= Rf°∩Sf°	ff°f=f	f°ff°=f°
f°· ⊣ f·	(A⇸C) → (B⇸C)	ff°	𝟙⊑ff°	f°f⊑𝟙	f°(X∪Y)= f°X∪f°Y	f(X∩Y)= fX∩fY	ff°f=f	f°ff°=f°
i ⊣ E	Map ↪ Rel	P	{·}:A⇸P A	∋:P B⇸B	—	—	Λ(∋)=𝟙	Λ(R)∋=R
·∋ ⊣ Λ	Map(A,P B) → (A⇸B)	𝟙	Λ(f∋)=f	Λ(R)∋=R	—	—	Λ(f∋)∋ =f∋	Λ(Λ(R)∋) =Λ(R)
Λ ⊣ ·∋	(A⇸B) → Map(A,P B)	𝟙	Λ(R)∋=R	Λ(f∋)=f	—	—	Λ(Λ(R)∋) =Λ(R)	Λ(f∋)∋ =f∋

§1.2. Composing adjunctions

(1.2a)

$$\frac{X}{f(R/T)} \underset{f^\circ\cdot\dashv f\cdot}{\Longleftrightarrow} \frac{f^\circ\ X}{R/T} \underset{\cdot T\dashv /T}{\Longleftrightarrow} \frac{(f^\circ\ X)\ T}{R}$$

f a map

$$\underset{\text{associativity}}{\Longleftrightarrow} \frac{f^\circ(X\ T)}{R} \underset{f^\circ\cdot\dashv f\cdot}{\Longleftrightarrow} \frac{X\ T}{f\ R} \underset{\cdot T\dashv /T}{\Longleftrightarrow} \frac{X}{(f\ R)/T}$$

$$\underset{\text{indirect equality}}{\Longrightarrow} \frac{f(R/T)}{(f\ R)/T}$$

	·T	T·	·g	g°·	T∩	°	⟜◁	Δ
·S	R/(ST)= (R/T)/S	T\(R/S)= (T\R)/S	R/(Sg)= (Rg°)/S	g(R/S)= (gR)/S	—	(Y/S)°= S°\(Y°)	(RS)^= R^(S⊗𝟙)	(T_1∩T_2)/S= T_1/S∩T_2/S
S·	S\(R/T)= (S\R)/T	(TS)\R= S\(T\R)	S\(Rg°)= (S\R)g°	(g°S)\R= S\(gR)	—	(S\Y)°= Y°/S°	((S⊗𝟙)R)^= S R^	S\(T_1∩T_2)= S\$T_1$∩S\$T_2$
·f	R/(fT)= (R/T)f°	T\(Rf°)= (T\R)f°	(fg)°=g°f°	—	—	(fY)°=Y°f°	—	(T_1∩T_2)f°= T_1f°∩T_2f°
f°·	f(R/T)= (fR)/T	(Tf°)\R= f(T\R)	—	(fg)°=g°f°	—	(fY)°=Y°f°	—	f(T_1∩T_2)= fT_1∩fT_2
R∩	—	—	—	—	(R∩T)⇒Y= R⇒(T⇒Y)	(R⇒Y)°= R°⇒(Y°)	—	R⇒(T_1∩T_2)= (R⇒T_1)∩(R⇒T_2)
°	(Y/T)°= T°\(Y°)	(T\Y)°= Y°/T°	(gY)°=Y°g°	(gY)°=Y°g°	(T⇒Y)°= T°⇒(Y°)	Y°°=Y	(R^)°= (R°⊗𝟙)(𝟙⊗▷⊸) both bends: the ° section's display	(T_1∩T_2)°= T_1°∩T_2°
⟜◁	(RT)^= R^(T⊗𝟙)	((T⊗𝟙)R)^= T R^	—	—	—	(R^)°= (R°⊗𝟙)(𝟙⊗▷⊸) both bends: the ° section's display	—	—
∪	(X_1∪X_2)T= X_1T∪X_2T	T(X_1∪X_2)= TX_1∪TX_2	(X_1∪X_2)g= X_1g∪X_2g	g°(X_1∪X_2)= g°X_1∪g°X_2	T∩(X_1∪X_2)= (T∩X_1)∪(T∩X_2)	(X_1∪X_2)°= X_1°∪X_2°	—	—
⊥	⊥T=⊥	T⊥=⊥	⊥g=⊥	g°⊥=⊥	T∩⊥=⊥	⊥°=⊥	—	—

(1.2b)

§1.3. Adjoint triples

Beyond ∪ ⊣ Δ ⊣ ∩, which §1.1 already carries as two rows, the table holds one adjoint triple with content: ∃_f ⊣ f* ⊣ ∀_f along a map f, once on each side of the composite. f : A → B throughout this subsection.

·f ⊣ ·f° ⊣ /f°
f°· ⊣ f· ⊣ f\ (1.3a)

The first two links of each chain are the map shunting rules, and the third is §1.1′s division row read at a chosen divisor: ·S ⊣ /S at S := f°, and S· ⊣ S\ at S := f.

The third link is not the first over again. For a map f, /f collapses to ·f°; being a map is exactly what that collapse spends R/(f S) = (R/S) f°. — f° is not a map, so /f° does not collapse in turn, and the three operators stay distinct. On Rel(𝕀, A) = P A they are the image triple:

(1.3b)

operator	acts by	name
·f	S ↦ f[S]	direct image, ∃_f
·f°	T ↦ f^{-1}[T]	inverse image, f*
/f°	S ↦ {b : f^{-1}(b) ⊆ S}	∀_f

§1.1′s 𝓓 ⊣ ·⊤ is this same chain along the projection A⊗B → A, read through Rel(A, B) = P(A⊗B): 𝓓R = {(a,a) : ∃b. aRb}, A⊤ is the relation that ignores b altogether, and the third link is R ↦ 𝟙 ∩ R/⊤ = {(a,a) : ∀b. aRb}.

𝓓 ⊣ ·⊤ ⊣ 𝟙∩·/⊤ (1.3c)

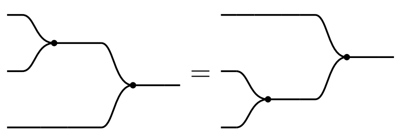
Three is where it stops, and one map breaks both ends. Take A = {a_1,a_2} and B = {b}: f[{a_1}] ∩ f[{a_2}] = {b} while f[{a_1} ∩ {a_2}] = ∅, so ·f does not preserve meets and has no left adjoint, and the same two subsets give ∀_f({a_1} ∪ {a_2}) = {b} against ∀_f{a_1} ∪ ∀_f{a_2} = ∅, so /f° does not preserve joins and has no right adjoint. A monic f restores the binary case and no more — ⊤f still reaches only im f, and ∀_f⊥ = B ∖ im f is still not ⊥. The chain extends only when f° is a map as well, and then /f° = ·f°° = ·f and it repeats forever. For an S with neither S nor S° a map there is no triple at all: ·S ⊣ /S is two links and stops.

The rows that chain forever say nothing by it: ° is an order-isomorphism of hom-posets, so it is adjoint to itself on both sides and ° ⊣ ° ⊣ ° has no end, and §1.1′s other row of that kind, ⟜◁ ⊣ ▷⊸, goes the same way.

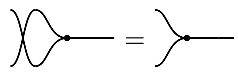
§2. Relations

Rel is a poset-enriched category with (⊗, °) where ⊗ is commutative cartisian product, and converse `°` ⊣ `°`, and C: `(▷ : A ⊗ A → A, ⟜ : 𝕀 → A)` a commutative monoid with `C°` ⊣ `C`. We write `C°` as `(◁ : A → A ⊗ A, ⊸ : A → 𝕀)` (2a)

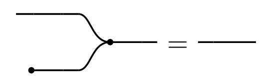
(2b)



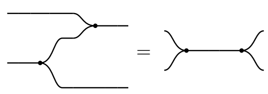
▷ associative



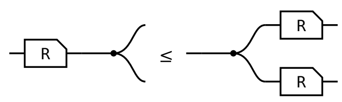
▷ commutative



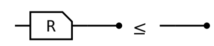
⟜ is its unit



Frobenius, one half — the other is its °



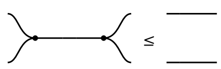
R◁ ≤ ◁(R⊗R)



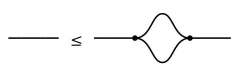
R⊸ ≤ ⊸

§2.1. ∀ object A, `(A, ◁, ⊸)` ⊣ `(A, ▷, ⟜)`

(2.1a)



▷ ◁ ≤ 𝟙



𝟙 ≤ ◁ ▷

⟜ ⊸ ≤ 𝟙

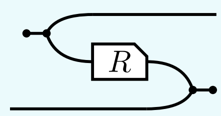
𝟙 ≤ ⊸ ⟜

The last of these is the only one that makes a picture **bigger**, and it is worth a name: **a wire is below the cut wire**. In `Rel` it reads `{(a, a)} ⊆ A × A` — cut a wire and its two ends stop having to agree, so cutting can only add pairs. It is the one weakening this calculus gives away for free.

§3. ° : $\mathcal{C}^{\mathbf{op}} \longrightarrow \mathcal{C}$ is a 2 functor

(3a)

° is primitive, part of the data the first section lists. It turns both of R's wires round, and the Frobenius structure DRAWS that — the picture and the formula below are R°, not its definition:



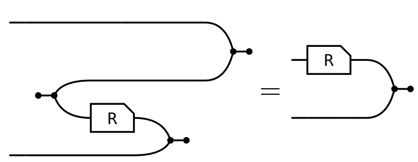
R° = (⟜◁ ⊗ 𝟙) (𝟙 ⊗ R ⊗ 𝟙) (𝟙 ⊗ ▷⊸)

where ⟜◁ : 𝕀 → a ⊗ a opens a pair of wires out of nothing and ▷⊸ : b ⊗ b → 𝕀 closes one, so the input of R° is where the output of R was. Taking that formula as the definition would now be circular: ◁ and ⊸ are ▷° and ⟜°. The laws of ° are that it is a contravariant 2-functor ° : $\mathcal{C}^{op}$ → $\mathcal{C}$:

(i) 𝟙° = 𝟙 (ii) (R S)° = S° R° (iii) (R ⊗ S)° = R° ⊗ S° (iv) R ≤ S implies R° ≤ S°

§3.1. The slide

The one rule (ii) and (iii) use, and each of them uses it twice. A converse facing a merge on the lower strand is the box itself, upright, on the upper one:



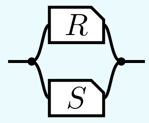
(3.1a)

R° is DEFINED as the bending of (R ⊗ 𝟙) ▷⊸, so the slide claims only that unbending it gives that back — and unbending undoes bending for every arrow. That is the snake above with a passenger: the ⟜◁ bends the a strand down and the ▷⊸ brings it back up, while the b strand rides through untouched. Nothing else is spent below.

§4. ∩ is a commutative idempotent monoid on every hom-set

(4a)

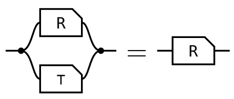
The **meet** of `R, S : a → b`, the paper's **convolution**, is `R ∩ S := ◁ (R ⊗ S) ▷` — copy the input, run `R` and `S` on the two copies, merge the results — so what comes out is what both of them do.



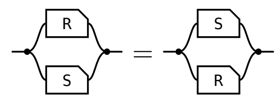
transcribed: a definition has no statement to export

On every hom-set it is associative, commutative and idempotent, with unit the maximal arrow `T = ⊸ ⟜` (the paper's Lemma 4.11).

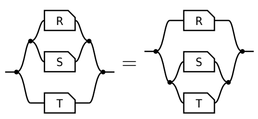
(4b)



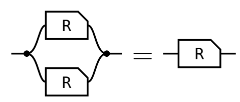
unit: one half of `T` per end — the merge's unit law absorbs the ⟜, the copy's counit law the ⊸



commutative: σ crosses `R ⊗ S` by naturality and is absorbed by cocommutativity and commutativity



associative: coassociativity and associativity; ⊗ re-brackets for nothing, being strict here

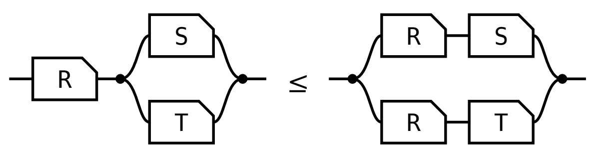


idempotent: the one that is not bookkeeping — the lax copy law is the whole of it, worked in allegory2

So ≤ is the order this monoid induces. `R ∩ S ≤ R` comes from the unit, and idempotency turns anything under both `S` and `T` into something under `S ∩ T`, since `R = R ∩ R ≤ S ∩ T`.

And one law relating ∩ to composition, which is **not** an equation:

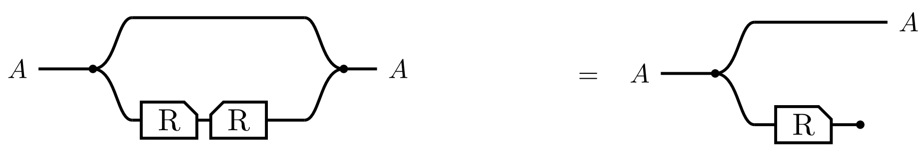
(4c)

semi-distributivity, and what supplies it	picture
`R (S ∩ T) ⊑ R S ∩ R T` — the lax copy law. Equality exactly when `R` is single valued: the Maps section's `F (R ∩ S) = F R ∩ F S`.	

§5. Domain and range

> The **domain** `Dom R ≜ 𝟙 ∩ R R°` and the **range** `Ran R ≜ Dom R°`. (5a)

(5b)



▷ lands the return leg back on the value ◁ handed out,
so `R°` ▷ cuts to ⊸ — Frobenius

Running `R` and throwing the result away leaves only the fact that `R` could fire, and `Ran R` is the same picture with the box mirrored. In `Rel` both steps are `{(a,a) : ∃b. a R b}`.

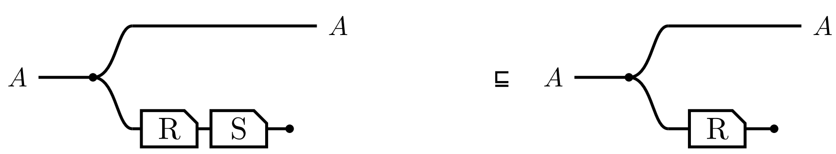
(5c)

`Dom R ≜ 𝟙 ∩ R R°`
`Dom R ⊑ A` ⟺ `R ⊑ A R`, for `A` coreflexive
`Dom (R S) ⊑ Dom R`
`Dom (R ∩ S) = 𝟙 ∩ S R°`
`R` entire ⟺ `Dom R = 𝟙` ⟺ `𝟙 ⊑ R R°`
`R` simple ⟺ `R° R ⊑ 𝟙`
`R` a map ⟺ `R` entire and simple
`R, S` entire ⟹ `R S` entire — likewise simple, likewise maps
`R S` entire ⟹ `R` entire

§5.1. Sliding the discard

`Dom (R S) ⊑ Dom R`, and a single glyph for `Dom` would have nothing to slide: with the box and the discard drawn apart, the law is one dot walking back along the lower strand.

(5.1a)



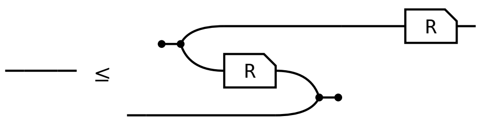
`S ⊸ ⊑ ⊸`, the lax axiom for ⊸ in the first section
— the discard slides back past `S`

Equality is `S` entire, which is the same picture read as `Dom R = 𝟙` ⟺ `R` entire.

§6. Maps

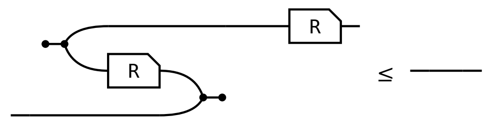
Every arrow is lax for ◁ and for ⊸ by axiom, and for ▷ and ⟜ by taking ° of those two. All four hold for **every** arrow, their REVERSES do not, and each reverse holding is a property of the arrow. The right-hand column is the **adjoint** form, which is the definition used here because it is literally the allegory's `Simple` and `Entire`; that the two forms agree is a separate theorem, not proved here.

(6a)



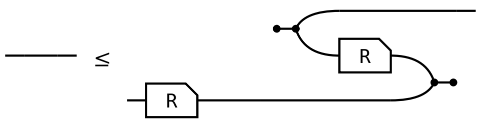
surjective

$\mathbb{1} \sqsubseteq R^\circ\ R$, that is, R° entire.



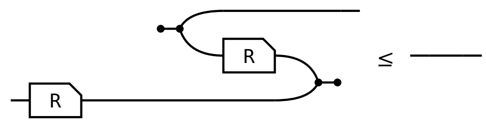
single valued

$R^\circ\ R \sqsubseteq \mathbb{1}$



entire

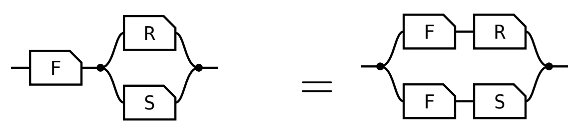
$\mathbb{1} \sqsubseteq R\ R^\circ$. With **single valued**, a **map**.



injective

$R\ R^\circ \sqsubseteq \mathbb{1}$

(6b)

law	picture
$F\ (R \cap S) = F\ R \cap F\ S$ F single valued F = people x may ask, R = admires a, S = admires b. Single valued means one person, so the sides agree.	 one person who admires both a at A, b at B

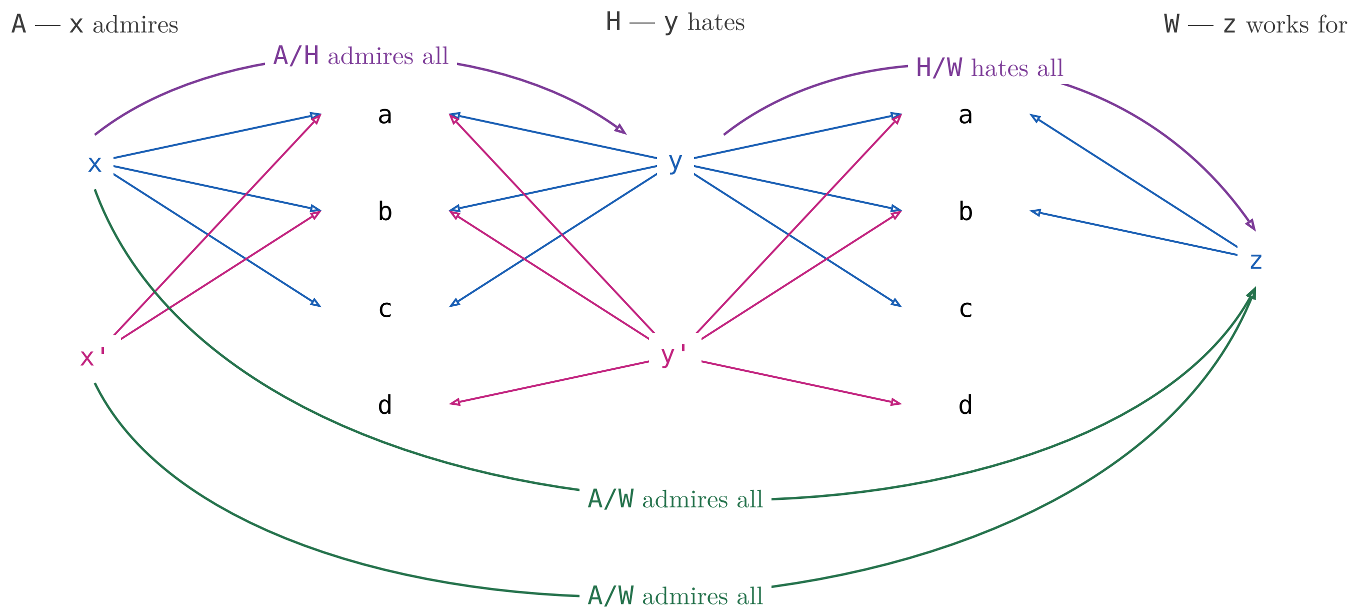
§7. `/` is all of

(7a)

```
x (R/S) y  ⟺  ∀p. y S p → x R p
x (S\R) y  ⟺  ∀p. p S x → p R y
/ compares images: S(y) ⊆ R(x).   \ compares preimages: S°(x) ⊆ R°(y).
example: A admires, H hates, W works for
x (A/H) y — x admires everyone y hates.
x (H\A) y — everyone who hates x admires y.
```

§7.1. `(R/S)(S/W) ⊑ R/W`

(7.1a)



(7.1b)

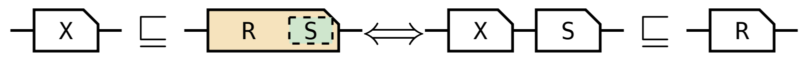
`x (A/H) y` — `x` admires everyone `y` hates
`y (H/W) z` — `y` hates everyone `z` works for
`x (A/W) z` — `x` admires everyone `z` works for

`(A/H)(H/W)` is a path: `x` → `y` → `z`, and that is all of it. `A/W` also holds of `x'`, who admires everyone `z` works for — but `x'` does not admire everyone anybody hates, so nothing composes to it. The missing path is exactly the strictness of `(R/S)(S/W) ⊑ R/W`.

(7.1c)

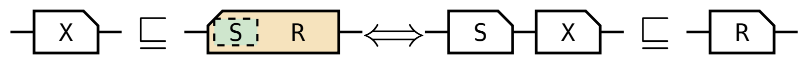
$X \sqsubseteq R/S \iff X\ S \sqsubseteq R$

X is any x-to-y pairing; one that only pairs x with a y such that x admires everyone y hates lies inside R/S, and R/S is the largest such.



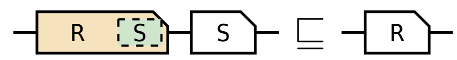
$X \sqsubseteq S\backslash R \iff S\ X \sqsubseteq R$

The mirror — divide on the left when x comes first.



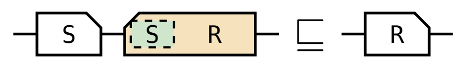
$(R/S)\ S \sqsubseteq R$

There is a y such that x admires everyone y hates, and p is one of the people y hates — then x admires p too. Strict at $S = \varnothing$: R/S is everyone, $(R/S)\ S = \varnothing$.



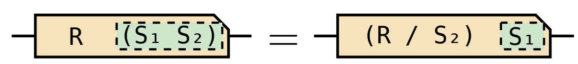
$S\ (S\backslash R) \sqsubseteq R$

The mirror.



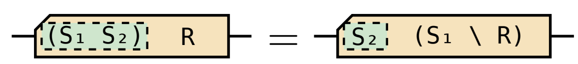
associate: $R/(S_1\ S_2) = (R/S_2)/S_1$

A friend's enemy is two hops: divide by the far end first.



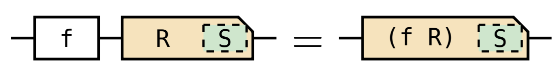
$(S_1\ S_2)\backslash R = S_2\backslash(S_1\backslash R)$

The mirror.



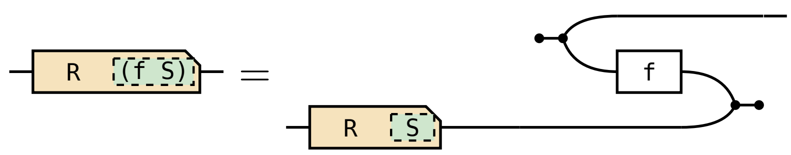
maps: $f\ (R/S) = (f\ R)/S$

Rename x before or after dividing — the licence to write $f\ R\ /\ S$.



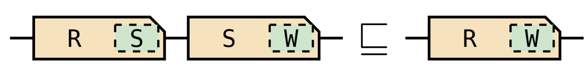
$R/(f\ S) = (R/S)\ f°$

Rename y: a map leaves a denominator as $f°$ outside the box.



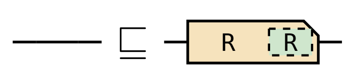
$(R/S)(S/W) \sqsubseteq R/W$

Someone who admires all of a hate-set that already covers everyone z works for admires those people too.



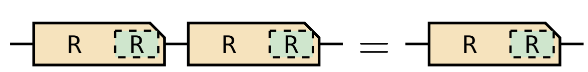
$\mathbb{1} \sqsubseteq R/R$

R/R runs admirer to admirer: each admires everyone they admire. Strict: two people who each admire only a and b admire each other's idols too, and still stay two people.



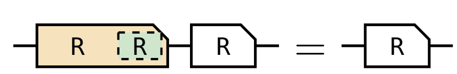
$(R/R)(R/R) = R/R$

R/R is the preorder **admires at least as much as**, and a preorder is idempotent. Freyd writes $\sqsubseteq$; with $\mathbb{1} \sqsubseteq R/R$ above it is an equality.



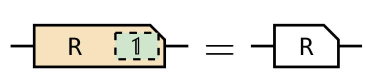
$(R/R)\ R = R$

Reaching p through someone whose idols x fully admires is reaching p directly, since x admires their own idols.



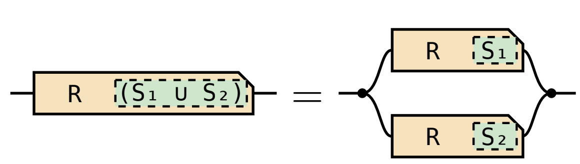
$R/\mathbb{1} = R$

Dividing by $\mathbb{1}$: p's set is just $\{p\}$, so admiring all of it is admiring p.



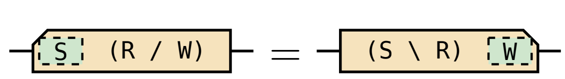
$R/(S_1 \cup S_2) = R/S_1 \cap R/S_2$

Admiring a combined hate-set is admiring each set in full.



$S\backslash(R/W) = (S\backslash R)/W$

Which is why $S\backslash R/W$ needs no bracket.



Fifteen laws, fifteen pictures, and not one shows a generator: $\cap$, $\cup$, $^\circ$ and composition are what the Frobenius generators build, and $/$ is none of those — it is posited, with nothing to unfold.

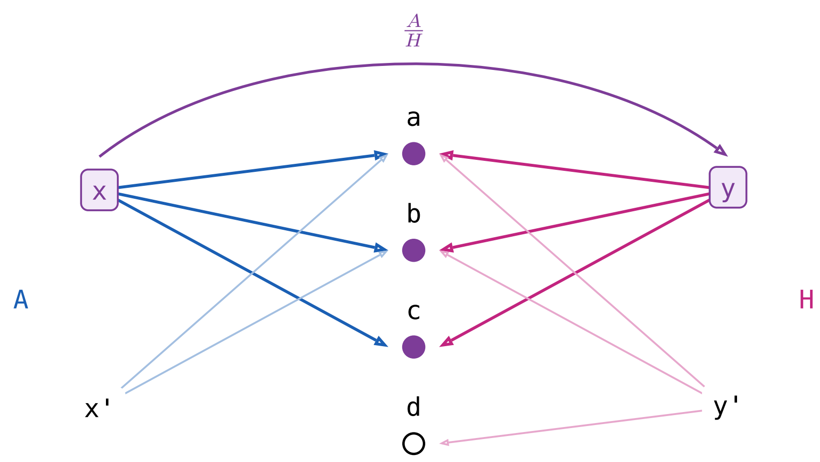
§8. $\frac{R}{S}$

$\frac{R}{S}$ ≜ `(R/S) ∩ (S/R)°`. In `Rel` `x` and `y` has the same image: `∀p. (x R p ⟺ y S p)` (8a)

`x (A/H) y` — `x` admires everyone `y` hates (8b)
`x ((H/A)°) y` — `x` admires only people `y` hates
`x ((A/H) ∩ (H/A)°) y` — `x` admires only and all whom `y` hates

`x` admires exactly whom `y` hates: `/` is **all of**, $\frac{R}{S}$ is **only all of**.

(8c)



(8d)

law	the reading
$X \sqsubseteq \frac{R}{S} \iff XS \sqsubseteq R$ and `X° R ⊑ S`	`X` may pair `x` with `y` only when `x` admires exactly whom `y` hates. Both halves must typecheck, so the operation is **partial**.
$\left(\frac{R}{S}\right)^{\circ} = \frac{S}{R}$	Matching is symmetric.
$\frac{R}{S}\frac{S}{W} \sqsubseteq \frac{R}{W}$	And transitive.
$\frac{R}{S}S \sqsubseteq R$	$(\exists y. x(\frac{R}{S})y \wedge ySp) \to xRp$ `x only admires whom y hates` $\frac{R}{S}S = \mathrm{Dom}(\frac{R}{S})R$

$\frac{R}{R}R = R$	$(\exists y. x(\frac{R}{R})y \wedge yRp) \iff xRp$
	`x and y admire the same people`
	`y = x always qualifies (𝟙 ⊑ R%R below)`

$\mathbb{1} \sqsubseteq \frac{R}{R}$	$x(\frac{R}{R})y$ if `x` and `y` admires the same peoples.
$\left(\frac{R}{R}\right)^2 = \frac{R}{R}$	So the relation **admires the same people** is an equivalence relation.
$X \sqsubseteq \frac{R}{R} \iff XR \sqsubseteq R$, for symmetric `X`	The largest symmetric arrow that leaves `R` alone.
$\frac{R}{\mathbb{1}}$ is the **simple part** of `R`	The people who admire exactly one person and nobody else. It equals `R` only when `R` is simple, unlike `R/𝟙 = R`.
`Dom` $\frac{R}{S}$ `= 𝟙 ∩ (R/S)(S/R)`	Its domain is the **domain of simplicity** of `R`.

§9. Freyd's Power allegories

A **power allegory** is a division allegory with one unary operation on arrows, ∋ **epsiloff**, subject to (9a)

$$\ni_R\Box = R\Box, \quad \ni_R = \ni_{R\,\Box}$$
$$\mathbb{1} \sqsubseteq (R/\ni_R)(\ni_R/R) \qquad \ni_R \text{ is } \textbf{thick}$$
$$(\ni_R/\ni_R) \cap (\ni_R/\ni_R)^\circ \sqsubseteq \mathbb{1} \qquad \ni_R \text{ is } \textbf{straight}$$

`R □` is `R`'s target, an identity arrow. For `R : A → B` write `∋ : P B → B`, dropping the subscript.

(9b)

law	the reading
$\ni_R\Box = R\Box, \quad \ni_R = \ni_{R\,\Box}$	∋ has the same target as `R`, and replacing `R` by the identity at that target leaves it unchanged: one ∋ per object, not per arrow.
∋ is **thick**	**Comprehension**: every `x` has a set of exactly the people `x` admires. Equivalently every `R` factors as a map followed by ∋.
∋ is **straight**, that is $\frac{\ni}{\ni} \sqsubseteq \mathbb{1}$	**Extensionality**: two sets with the same people are the same set.
`Λ(R)` ≜ $\frac{R}{\ni}$ `: A → P B`, for `R : A → B`	convert a relation to a function. `a Λ(R) = {b|a R b}`, image of a
`Λ(R)` is simple	∋ is straight, and dividing by a straight arrow is simple. At most one set per `x`.
`Λ(R)` is entire ⟺ ∋ thick	`Dom` $\frac{R}{\ni}$ `= 𝟙 ∩ (R/∋)(∋/R)`, the domain row above. At least one set per `x`, so `Λ(R)` is a **map**.
`Λ(R) ∋ = R`	Drawn in §10, where it is the counit's cancellation.
`Λ(R)` is the only map with `Λ(R) ∋ = R`	Two maps naming the same people name the same set — extensionality again.
`F ⊑ Λ(F ∋)`, `F` simple	A partial choice of sets is inside the total one.
`P A` ≜ source of ∋, the **power-object**	Notation, not a law: ∋ goes `P A → A`, and `P A` is simply a name for where it starts. Every subset of `A`.
`{·}` ≜ `Λ(𝟙)`, the **singleton map**, monic	The one-person set. `Λ(𝟙)Λ°(𝟙) ⊑` $\frac{\mathbb{1}}{\ni}\frac{\ni}{\mathbb{1}} \sqsubseteq \frac{\mathbb{1}}{\mathbb{1}} \sqsubseteq \mathbb{1}$.
`Λ(∋) = 𝟙`	Make the set of a set, then read it back one level down. Straightness itself, and one of the two snake laws of §10; the other is `{·} ∋ = 𝟙`.
fusion: `Λ(f R) = f Λ(R)`, `f` a map	Naturality of the unit, `f {·} = {·} (E(f))`, and that is the whole content of fusion. Drawn in §10.
`Λ(f) = f {·}`, `f` a map	Rename first or take singletons first — the fusion row above at `R = 𝟙`.
$\frac{R}{S} = \Lambda(R)\Lambda^\circ(S)$	`x` and `y` match exactly when `Λ(R)x` and `Λ(S)y` are one and the same set.
`E(R) ≜ Λ(∋ R)`	`E(R): P A → P B`, image of a set of A
`Λ(R) = {·} E(R)`	{ · }: x ↦ {x}
`Λ(S) E(R) = Λ(S R)`	absorption
`subset ≜ ∋/∋ : P A → P A`	`x subset y ⟺ ∀a. y ∋ a → x ∋ a`, that is `y ⊆ x`: diagram order reverses B&dM's argument order along with the arrow. It **models** inclusion between sets, where `⊑` is the allegory's own primitive comparing arrows (B&dM p. 106).

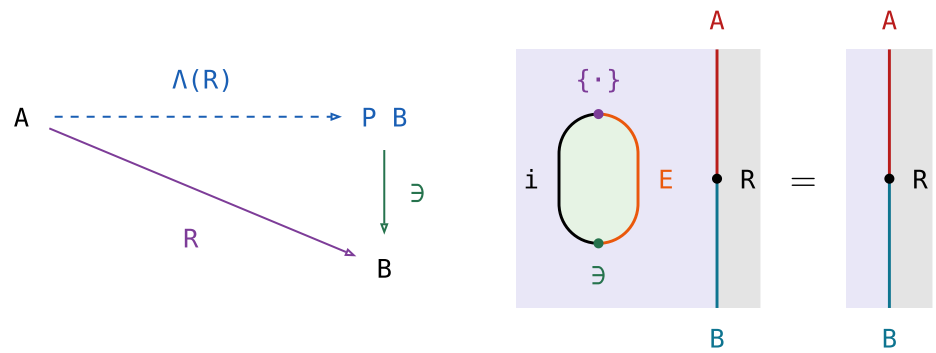
§10. `i ⊣ E`

`E(R) ≜ Λ(∋ R)` is Bird & de Moor's **existential image** (p. 105), and it is the right adjoint of the inclusion `i : Map(𝒜) ↪ 𝒜`: `Λ` is the transpose, so `𝒜(A, B) ≅ Map(A, P B)`.

(10a)

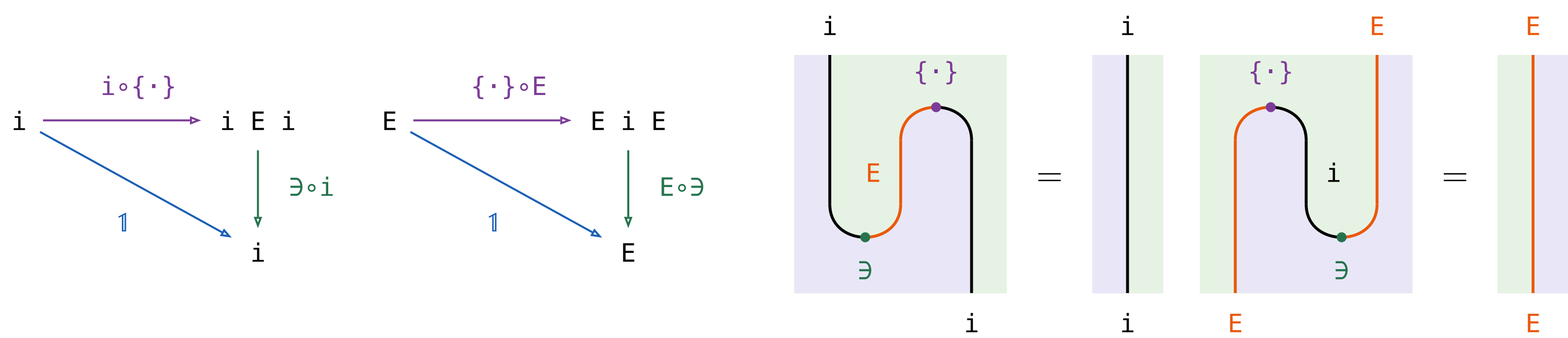
part of `i ⊣ E`	**type**	**the statement**
`∋` the counit, natural	`∋ : P B → B`	`E(R) ∋ = ∋ R`
`{·}` the unit, natural	`{·} : A → P A`	`f {·} = {·} E(f)`, for `f` a map
`Λ` the transpose	`Λ : (A → B) → Map(A, P B)`	`Λ(R) ∋ = R`, and `Λ(R)` the only map with it
`·∋` the transpose back	`·∋ : Map(A, P B) → (A → B)`	`{·} ∋ = 𝟙` and `Λ(∋) = 𝟙`, the two triangles

(10b)



`Λ(R) ∋ = R`
and `Λ(R)` is the only **map** with that property

(10c)

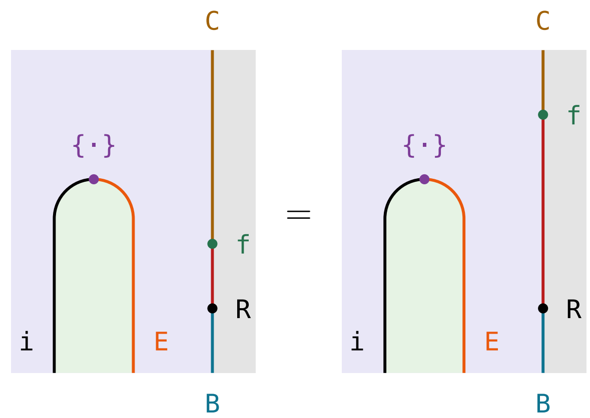


`{·} ∋ = 𝟙` `Λ(∋) = 𝟙`

§10.1. Fusion law

(10.1a)

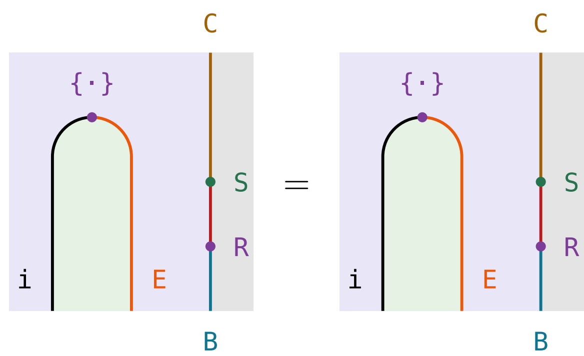
$$\begin{tikzcd} C \arrow[r, "f"] \arrow[dr, dashed, "\Lambda(f\ R)"'] & A \arrow[d, dashed, "\Lambda(R)"] \\ & P\ B \end{tikzcd}$$



`Λ(f R) = f Λ(R)`, `f` a map

(10.1b)

$$\begin{tikzcd} C \arrow[r, dashed, "\Lambda(S)"] \arrow[dr, dashed, "\Lambda(S\ R)"'] & P\ A \arrow[d, "E(R)"] \\ & P\ B \end{tikzcd}$$



`Λ(S R) = Λ(S) E(R)`, for `S : C → A` and `R : A → B`

(10.1c)

scan line	**the object on it**	**the bead just passed**
the top edge	`C`	—
between `{·}` and `S`	`i E C = P C`	`{·} : C → P C`
between `S` and `R`	`i E A = P A`	`S`, inside the pair, so `E(S) : P C → P A`
the bottom edge	`i E B = P B`	`R`, likewise `E(R) : P A → P B`

One string of beads, two groupings:

- The first two, then the third: `{·} E(S) = Λ(S) : C → P A` followed by `E(R) : P A → P B`.
- All three at once: `{·} E(S) E(R) = {·} E(S R) = Λ(S R) : C → P B`.

Absorption holds for every `S`, so fusion is its case `S := f`, plus one step: absorption there reads `Λ(f R) = Λ(f) E(R)`, and turning `Λ(f) E(R)` into `f Λ(R)` needs `Λ(f) = f {·}`, which is (10.1a) at `R = 𝟙`. That step is the unit's naturality, and it is all fusion adds.

§10.2. `Λ(R) = (R/∋) ∩ (∋/R)°`

(10.2a)

$$\underset{f,\ g\ \text{maps}}{\frac{g^\circ\ f}{\Lambda(R)}} \xLeftrightarrow[f^\circ\cdot\dashv f\cdot]{} \frac{f}{g\ \Lambda(R)} \xLeftrightarrow[\text{fusion}]{} \frac{f}{\Lambda(g\ R)} \xLeftrightarrow[\cdot\ni\dashv\Lambda]{} \frac{f\ \ni}{g\ R}$$

$$\xLeftrightarrow[\text{antisymmetry}]{} \begin{matrix}\frac{f\ \ni}{g\ R} \\ \frac{g\ R}{f\ \ni}\end{matrix} \xLeftrightarrow[f^\circ\cdot\dashv f\cdot]{} \begin{matrix}\frac{g^\circ\ f\ \ni}{R} \\ \frac{f^\circ\ g\ R}{\ni}\end{matrix} \xLeftrightarrow[\cdot S\dashv/S]{} \begin{matrix}\frac{g^\circ\ f}{R/\ni} \\ \frac{f^\circ\ g}{\ni/R}\end{matrix}$$

$$\xLeftrightarrow[{}^\circ,\ \text{meet}]{} \frac{g^\circ\ f}{(R/\ni)\ \cap\ (\ni/R)^\circ}$$

§11. Relator

> Every hom-set of an allegory is a poset, so an allegory is a **locally posetal 2-category**: the 2-cell from $\mathtt{R}$ to $\mathtt{S}$ IS $\mathtt{R} \sqsubseteq \mathtt{S}$. A **relator** $\mathtt{F} : \mathcal{C} \to \mathcal{D}$ is a 2-functor between allegories: (11a)
>
> $$\mathtt{F}(\mathbb{1}) = \mathbb{1} \qquad \mathtt{F}(\mathtt{R}\ \mathtt{S}) = \mathtt{F}(\mathtt{R})\ \mathtt{F}(\mathtt{S}) \qquad \mathtt{R} \sqsubseteq \mathtt{S} \Longrightarrow \mathtt{F}(\mathtt{R}) \sqsubseteq \mathtt{F}(\mathtt{S})$$
>
> Preserving $^\circ$ is **not** asked for — $^\circ$ is an identity-on-objects involution $\mathcal{C}^{op} \to \mathcal{C}$, no part of the 2-category.

(11b)

the statement	**B&dM**
For $\mathtt{f}$ a map, $\mathtt{F(f)}$ is a map and $\mathtt{F}(\mathtt{f}^\circ) = \mathtt{F}(\mathtt{f})^\circ$.	Lemma 5.1
Over a **tabular** allegory a functor is a relator $\Longleftrightarrow$ it preserves $^\circ$.	Theorem 5.1
$\mathtt{F}(\mathtt{R}^\circ) = \mathtt{F}(\mathtt{R})^\circ$ for every $\mathtt{R}$, so $\mathtt{F}(\mathtt{R})^\circ$ needs no bracket.	after Theorem 5.1, p. 113
Two relators agreeing on maps are equal.	Corollary 5.1
$\mathtt{F}(\mathtt{X} \cap \mathtt{Y}) = \mathtt{F}(\mathtt{X}) \cap \mathtt{F}(\mathtt{Y})$ for $\mathtt{X},\ \mathtt{Y}$ coreflexive.	Ex 5.2
$\mathtt{F}(\mathtt{R} \cap \mathtt{S}) \sqsubseteq \mathtt{F}(\mathtt{R}) \cap \mathtt{F}(\mathtt{S})$, and strictly.	Ex 5.2, the restriction
$\mathtt{F}(\mathrm{Dom}\ \mathtt{R}) = \mathrm{Dom}(\mathtt{F}(\mathtt{R}))$ for $\mathtt{F}$ preserving $^\circ$.	—

The **power relator** $\mathtt{P}$ — $\mathtt{x}\ \mathtt{P(R)}\ \mathtt{y} \iff (\forall \mathtt{a} \in \mathtt{x}.\ \exists \mathtt{b} \in \mathtt{y}.\ \mathtt{a}\ \mathtt{R}\ \mathtt{b}) \wedge (\forall \mathtt{b} \in \mathtt{y}.\ \exists \mathtt{a} \in \mathtt{x}.\ \mathtt{a}\ \mathtt{R}\ \mathtt{b})$ — is where the fourth is strict: for $\mathtt{R} = \{(\mathtt{a}_1,\mathtt{b}_1),\ (\mathtt{a}_2,\mathtt{b}_2)\}$ and $\mathtt{S} = \{(\mathtt{a}_1,\mathtt{b}_2),\ (\mathtt{a}_2,\mathtt{b}_1)\}$ the pair $(\{\mathtt{a}_1,\mathtt{a}_2\},\ \{\mathtt{b}_1,\mathtt{b}_2\})$ is in $\mathtt{P(R)} \cap \mathtt{P(S)}$, while $\mathtt{R} \cap \mathtt{S} = \varnothing$.

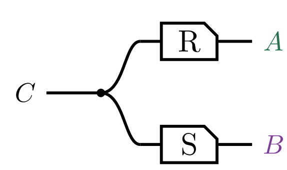
§12. Fork $\langle\mathtt{R},\mathtt{S}\rangle$

> The **fork** of $\mathtt{R} : \mathtt{C} \to \mathtt{A}$ and $\mathtt{S} : \mathtt{C} \to \mathtt{B}$ is $\langle\mathtt{R},\mathtt{S}\rangle \triangleq \mathtt{R}\ \pi_1^\circ \cap \mathtt{S}\ \pi_2^\circ$, where $(\pi_1,\ \pi_2)$ is the tabulation of $\top$. (12a)

$$\langle\mathtt{R},\mathtt{S}\rangle\ \pi_1 = (\mathrm{Dom}\ \mathtt{S})\ \mathtt{R} \qquad \langle\mathtt{R},\mathtt{S}\rangle\ \pi_2 = (\mathrm{Dom}\ \mathtt{R})\ \mathtt{S} \tag{12b}$$

(12c)

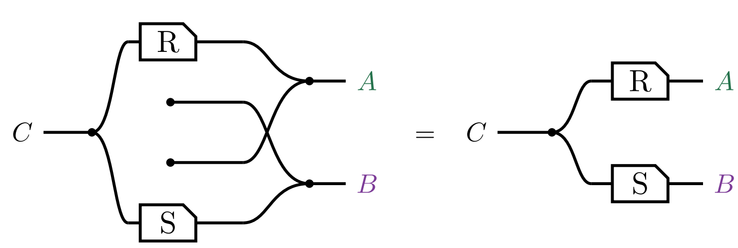
$$\begin{tikzcd}
 & A & \\
C \arrow[ur, "\mathsf{R}"] \arrow[rr, dashed, "{\langle R, S\rangle}"] \arrow[dr, "\mathsf{S}"'] & \arrow[r, phantom, "\sqsubseteq"] & A \times B \arrow[ul, "\pi_1"'] \arrow[dl, "\pi_2"] \\
 & B &
\end{tikzcd}$$



A domain is coreflexive, so $\langle\mathtt{R},\mathtt{S}\rangle\ \pi_1 \sqsubseteq \mathtt{R}$, with equality exactly when $\mathtt{S}$ is entire; for maps both triangles commute and $\langle\mathtt{f},\mathtt{g}\rangle$ is unique. In $\mathtt{Rel}$, $\mathtt{c}\ \langle\mathtt{R},\mathtt{S}\rangle\ (\mathtt{a},\mathtt{b})$ iff $\mathtt{c}\ \mathtt{R}\ \mathtt{a}$ and $\mathtt{c}\ \mathtt{S}\ \mathtt{b}$ — copy $\mathtt{c}$, then $\mathtt{R}$ on one strand and $\mathtt{S}$ on the other, which is $\lhd\ (\mathtt{R} \otimes \mathtt{S})$ on the right.

No $^\circ$ survives the translation. $\pi_1 = \mathbb{1} \otimes \multimap$ discards the second component, so $\pi_1^\circ = \mathbb{1} \otimes \multimapinv$ **creates** one out of nothing, and $\cap$ is copy, run both, merge. Draw that and the created strands — the two dots with no left end, and the crossing they force — are merged against real ones, which is the monoid's unit law:

(12d)

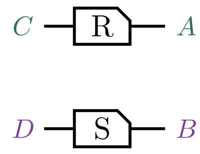


$\multimapinv\ \rhd = \mathbb{1}$ on each half

§12.1. Relational product $\mathtt{R} \times \mathtt{S}$

> $\mathtt{R} \times \mathtt{S} \triangleq \langle\pi_1\ \mathtt{R},\ \pi_2\ \mathtt{S}\rangle$, a relator in each argument but no longer a categorical product. (12.1a)

$$\begin{tikzcd} C \arrow[r, "R"] \arrow[dr, phantom, "\sqsupseteq"] & A \\ C \times D \arrow[u, "\pi_1"] \arrow[d, "\pi_2"'] \arrow[r, dashed, "R \times S"] \arrow[dr, phantom, "\sqsupseteq"] & A \times B \arrow[u, "\pi_1"'] \arrow[d, "\pi_2"] \\ D \arrow[r, "S"] & B \end{tikzcd} \tag{12.1b}$$

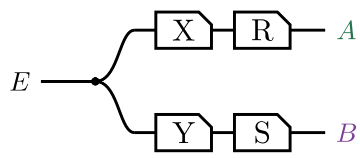


Right-then-up is $(R \times S)\ \pi_1$, up-then-right is $\pi_1\ R$, and $(R \times S)\ \pi_1 \sqsubseteq \pi_1\ R$, equality when S is entire. In Rel, $(c,d)\ (R \times S)\ (a,b)$ iff $c\ R\ a$ and $d\ S\ b$ — two strands side by side, no copy dot: $R \times S = R \otimes S$.

§12.2. Absorption

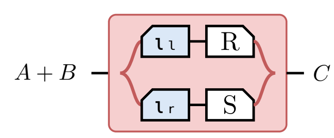
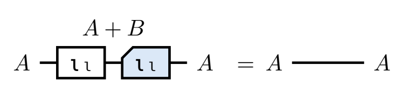
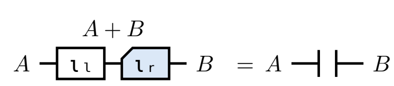
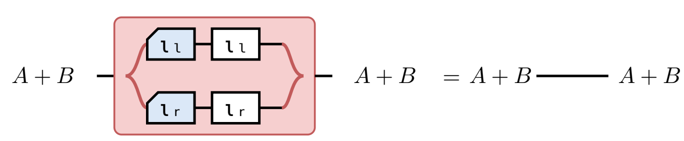
For $X : E \to C$ and $Y : E \to D$, $\langle X,Y\rangle\ (R \times S) = \langle X\ R,\ Y\ S\rangle$. Both sides are this picture:

(12.2a)



§13. Coproduct $[R,S]$

(13a)

the statement	picture
$[R,S] \triangleq \iota_l{}^\circ\ R \cup \iota_r{}^\circ\ S$ The tape is the union — a particle entering at $A + B$ takes exactly one branch — and the two mirrored boxes are what makes the branches disjoint.	
$[R,S] = [\Lambda(R),\ \Lambda(S)]\ \ni$	
$R + S \triangleq [R\ \iota_l,\ S\ \iota_r]$	
$\iota_l\ [R,S] = R$, $\iota_r\ [R,S] = S$, and $[R,S]$ is the only such arrow	
$\iota_l\ \iota_l{}^\circ = 𝟙 = \iota_r\ \iota_r{}^\circ$	
$\iota_l\ \iota_r{}^\circ = \bot = \iota_r\ \iota_l{}^\circ$	
$\iota_l{}^\circ\ \iota_l \cup \iota_r{}^\circ\ \iota_r = 𝟙$	
$[U,V]^\circ\ [R,S] = U^\circ\ R \cup V^\circ\ S$	

§13.1. $[R,S] \triangleq [\Lambda(R),\ \Lambda(S)]\ \ni$

The universal-property row is not free: ι_l, ι_r were only ever asked to be a coproduct of **maps**. They stay one once every arrow is allowed because Λ sends an arrow $A \to C$ to a map $A \to P\ C$ reversibly, so the map coproduct can be applied underneath it. For any $T : A + B \to C$,

$$\boxed{\frac{\iota_l\ T}{R}}\ \boxed{\frac{\iota_r\ T}{S}} \overset{}{\underset{\cdot\ni\dashv\Lambda}{\Longleftrightarrow}} \boxed{\frac{\Lambda(\iota_l\ T)}{\Lambda(R)}}\ \boxed{\frac{\Lambda(\iota_r\ T)}{\Lambda(S)}} \underset{\text{fusion}}{\Longleftrightarrow} \boxed{\frac{\iota_l\ \Lambda(T)}{\Lambda(R)}}\ \boxed{\frac{\iota_r\ \Lambda(T)}{\Lambda(S)}} \tag{13.1a}$$

$$\xLeftrightarrow[\text{coproduct of maps}]{} \boxed{\dfrac{\Lambda(\mathsf{T})}{[\Lambda(\mathsf{R}),\ \Lambda(\mathsf{S})]}} \xLeftrightarrow[\cdot\ni\dashv\Lambda]{} \boxed{\dfrac{\mathsf{T}}{[\Lambda(\mathsf{R}),\ \Lambda(\mathsf{S})]\ \ni}}$$

$$\begin{tikzcd}
& A \arrow[dl, "\iota_l"'] \arrow[dr, "\Lambda(\mathsf{R})"] \arrow[drr, bend left, "\mathsf{R}"] & & \\
A+B \arrow[rr, dashed, "{[\Lambda(\mathsf{R}),\ \Lambda(\mathsf{S})]}"] & & PC \arrow[r, "\ni"] & C \\
& B \arrow[ul, "\iota_r"] \arrow[ur, "\Lambda(\mathsf{S})"'] \arrow[urr, bend right, "\mathsf{S}"'] & &
\end{tikzcd} \tag{13.1b}$$

Nothing here holds only up to $\sqsubseteq$: every triangle commutes on the nose, which is the difference from the fork above. The border spells $[\mathsf{R},\mathsf{S}] = [\Lambda(\mathsf{R}),\ \Lambda(\mathsf{S})]\ \ni$, and pushing $\ni$ into the union that $[\cdot,\cdot]$ on maps already is turns that back into the definition, $(\iota_l{}^\circ\ \Lambda(\mathsf{R}) \cup \iota_r{}^\circ\ \Lambda(\mathsf{S}))\ \ni\ = \iota_l{}^\circ\ \mathsf{R} \cup \iota_r{}^\circ\ \mathsf{S}$.

§14. The power relator `P(R)`

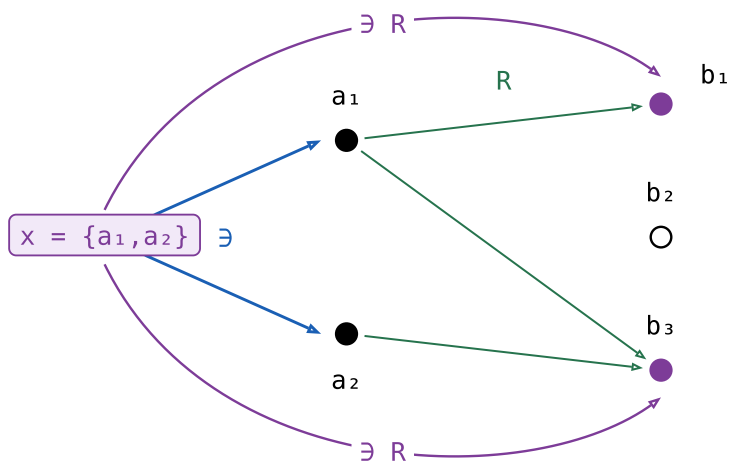
(14a)

For $R : A \to B$, $P(R) \triangleq ((\ni R)/\ni) \cap ((\ni R^\circ)/\ni)^\circ : P\,A \to P\,B$.

$x\ P(R)\ y \iff (\forall a \in x.\ \exists b \in y.\ a\ R\ b) \land (\forall b \in y.\ \exists a \in x.\ a\ R\ b)$

Every element of x is related by R to some element of y, and conversely.

(14b)

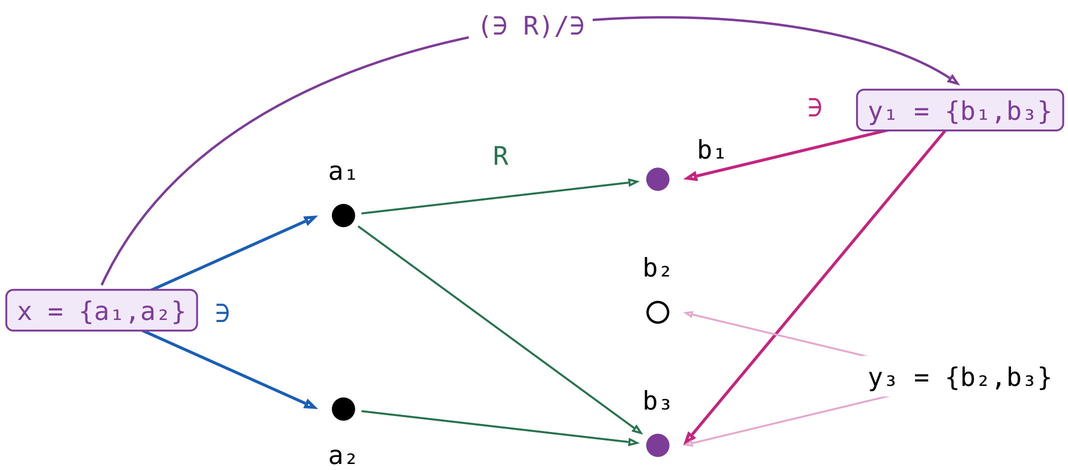


$\ni R : P\,A \to B \qquad x\ (\ni R)\ b \iff \exists a \in x.\ a\ R\ b$

one arc per element of B that x reaches, and b_2 gets none

Dividing by $\ni$ turns that into a relation between **sets**.

(14c)

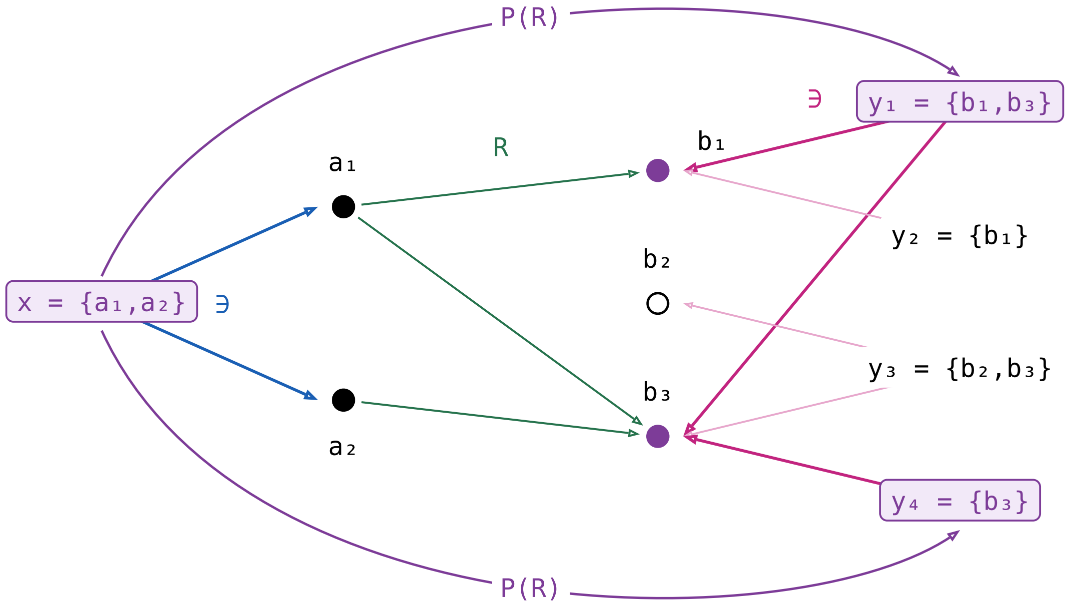


$(\ni R)/\ni : P\,A \to P\,B \qquad x\ ((\ni R)/\ni)\ y \iff \forall b \in y.\ \exists a \in x.\ a\ R\ b$

y_3 is rejected: it names b_2, which x does not reach

The other half of the meet is that clause with the sets swapped and R turned round: every element of x must reach y:

(14d)



$$((\ni\ R^\circ)/\ni)^\circ : P\ A \to P\ B \qquad x\ ((\ni\ R^\circ)/\ni)^\circ\ y \iff \forall a \in x.\ \exists b \in y.\ a\ R\ b$$

y_2 is rejected too: a_2's only image is b_3, which y_2 does not name

(14e)

law	**the reading**
$X \sqsubseteq P(R) \iff X \ni \sqsubseteq \ni R$ and $X^\circ \ni \sqsubseteq \ni R^\circ$	One containment, and the same one at R° — which is the definition read off the two divisions. Hence $P(R^\circ) = P(R)^\circ$, and $R \sqsubseteq S \implies P(R) \sqsubseteq P(S)$.
$P(\mathbb{1}) = \frac{\ni}{\ni} = \mathbb{1}$	The straightness axiom verbatim: extensionality **is** P's unit law.
$P(f) = \Lambda(\ni\ f) = \frac{\ni f}{\ni}$, for f a map	In Rel, $x\ P(f)\ y \iff y = \{f\ a \mid a \in x\}$. The half at f° says every $a \in x$ has its $f\ a$ on y; f has just the one image per a, so that already says y contains everything x reaches, which is the fraction's second half. For a map the two definitions coincide.
$P(R\ S) = P(R)\ P(S)$	$\sqsupseteq$ is the division cancellation laws. $\sqsubseteq$ is the one law in this section that is not a calculation: it needs a tabulation of $P(R\ S)$.

§15. Catamorphism

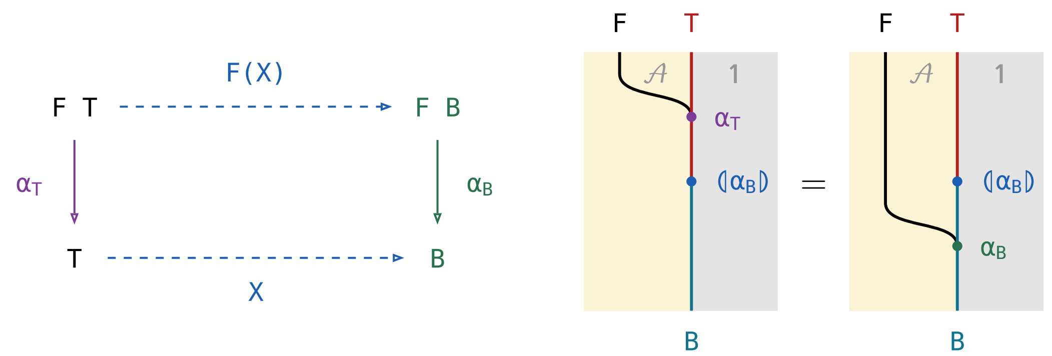
(15a)

> F a relator with an **initial algebra** $\alpha_T : F\ T \to T$ among the maps. For a relational algebra $\alpha_B : F\ B \to B$, the **catamorphism** $(\!|\alpha_B|\!) : T \to B$ is the unique arrow with $\alpha_T\ (\!|\alpha_B|\!) = F((\!|\alpha_B|\!))\ \alpha_B$. Explicitly, for a relation $R : F\ B \to B$, $(\!|R|\!) \triangleq (\!|\Lambda(F(\ni)\ R)|\!)\ \ni$, the inner $(\!|\cdot|\!)$ being the catamorphism of a **map** algebra.

An F-algebra is a **weakened** natural transformation. A transformation $F \Rightarrow 1\!\!1$ on $\mathcal{A}$ would need a component $F\ X \to X$ at every object and a commuting square at every arrow, but F-Algebra only need it works on T and B.

§15.1. The defining equation

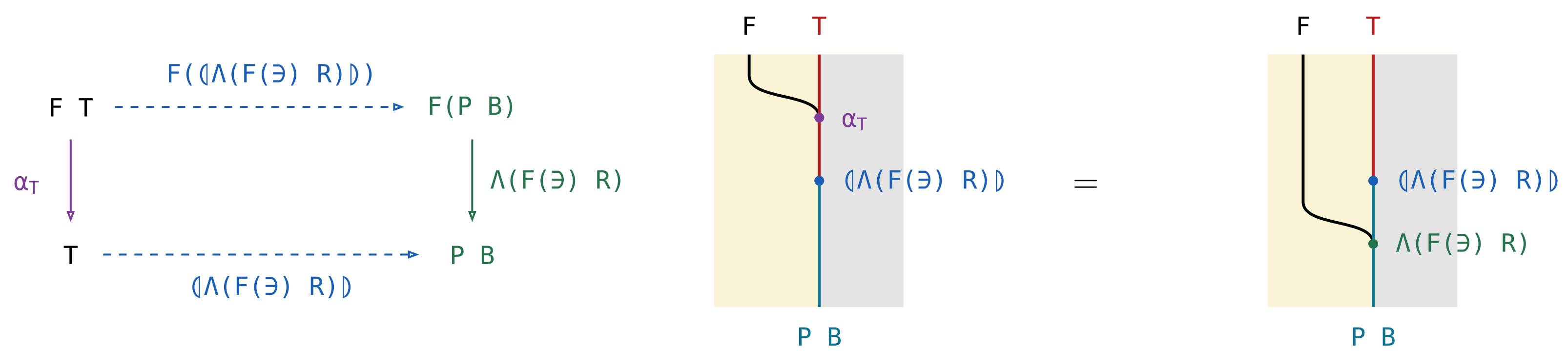
(15.1a)



$X = (\!|\alpha_B|\!) \iff \alpha_T\ X = F(X)\ \alpha_B$

§15.2. $(\!|R|\!) = (\!|\Lambda(F(\ni)\ R)|\!)\ \ni$

(15.2a)



$\alpha_T\ (\!|\Lambda(F(\ni)\ R)|\!) = F((\!|\Lambda(F(\ni)\ R)|\!))\ \Lambda(F(\ni)\ R)$

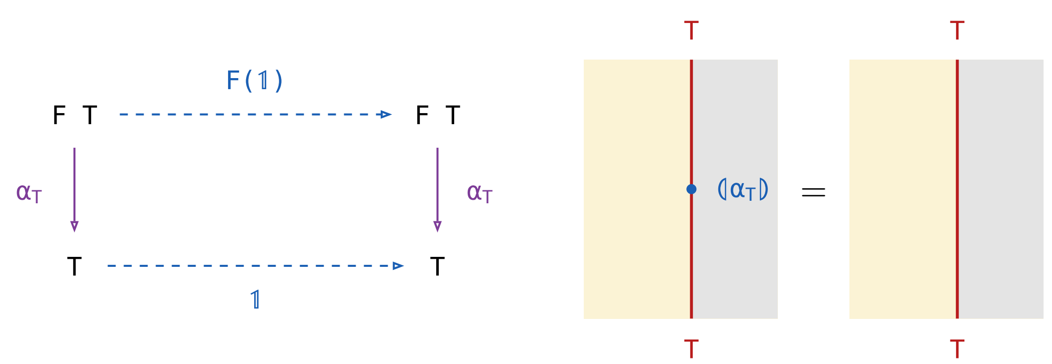
(15.2b)

$$\dfrac{\alpha_T\ X}{F(X)\ R} \xLeftrightarrow[\cdot\ni\dashv\Lambda]{} \dfrac{\Lambda(\alpha_T\ X)}{\Lambda(F(X)\ R)} \xLeftrightarrow[\cdot\ni\dashv\Lambda]{} \dfrac{\Lambda(\alpha_T\ X)}{\Lambda(F(\Lambda(X)\ \ni)\ R)}$$

$$\xLeftrightarrow[\text{relator, fusion twice}]{} \dfrac{\alpha_T\ \Lambda(X)}{F(\Lambda(X))\ \Lambda(F(\ni)\ R)} \xLeftrightarrow[\text{catamorphism of maps}]{} \dfrac{\Lambda(X)}{(\!|\Lambda(F(\ni)\ R)|\!)} \xLeftrightarrow[\cdot\ni\dashv\Lambda]{} \dfrac{X}{(\!|\Lambda(F(\ni)\ R)|\!)\ \ni}$$

§15.3. Reflection

(15.3a)



$(\!|\alpha_T|\!) = 1\!\!1$

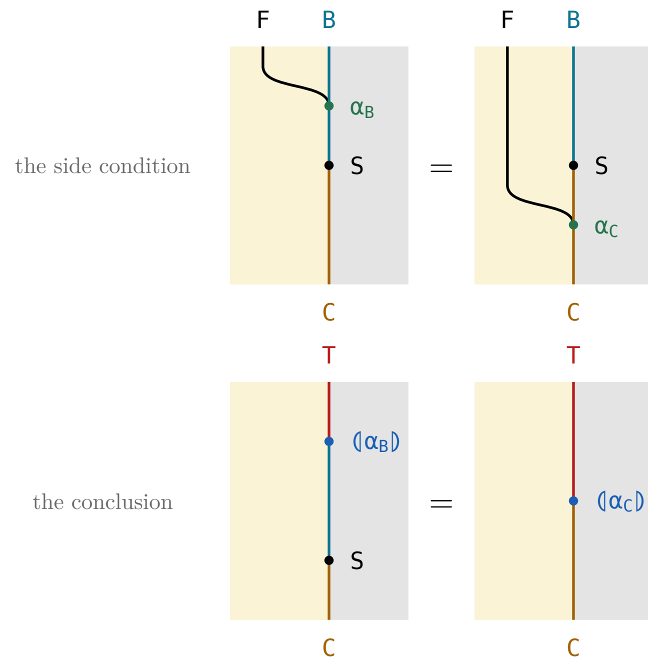
Taking a value apart with α_T and putting it straight back is doing nothing.

§15.4. Fusion

Fusion rewrites `⦇α_B⦈ S` through a second algebra α_C `: F C → C` along an arrow `S : B → C`.

(15.4a)

$$\begin{tikzcd}
F\,T \arrow[r, dashed, "F(⦇\alpha_B⦈)"] \arrow[d, "\alpha_T"'] & F\,B \arrow[r, "F(S)"] \arrow[d, "\alpha_B"] & F\,C \arrow[d, "\alpha_C"] \\
T \arrow[r, dashed, "⦇\alpha_B⦈"'] & B \arrow[r, "S"'] & C
\end{tikzcd}$$



$$⦇\alpha_B⦈\ S = ⦇\alpha_C⦈ \Longleftarrow \alpha_B\ S = F(S)\ \alpha_C$$

The left square is `⦇α_B⦈`'s own defining square and the right one is the side condition, so the outer rectangle says `⦇α_B⦈ S` satisfies the defining equation of `⦇α_C⦈` — and uniqueness finishes it. A fold followed by `S` has collapsed into a single fold, which is how an intermediate structure is got rid of.

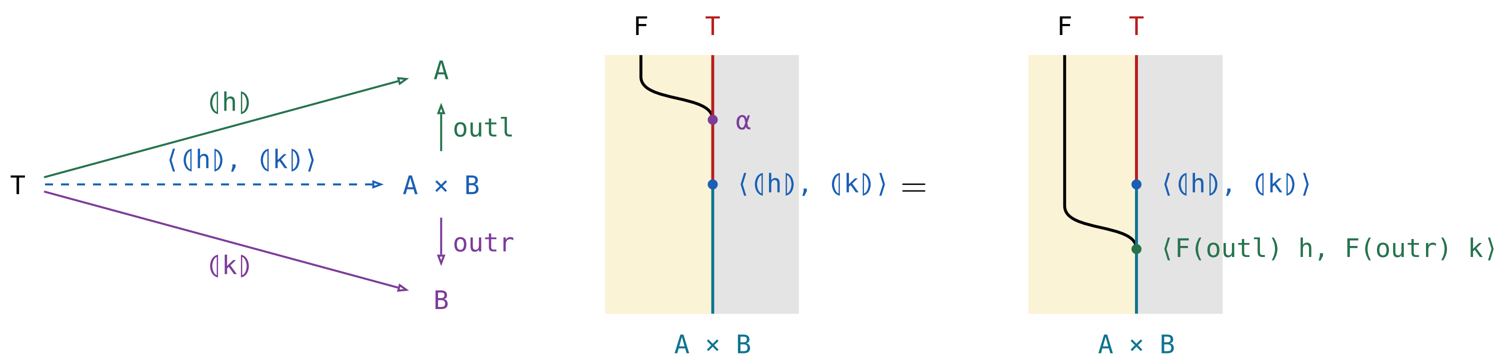
The side condition is (15.1a)'s picture again under $\alpha_T \mapsto \alpha_B$, $⦇\alpha_B⦈ \mapsto S$, $\alpha_B \mapsto \alpha_C$: both say that the bead between the two merges is a homomorphism of `F`-algebras.

§15.5. Catamorphism in $\mathcal{S}$et

(15.5a)

name	definition	note
`listr`	`listr A ::= nil | cons (A, listr A)`	The cons-lists over `A`, the datatype every row below folds (B&dM p. 55).
`sum`	`⦇[zero, plus]⦈`	`plus(a, b) = a + b`.
`length`	`⦇[zero, outr succ]⦈`	`outr` drops the head and keeps the count of the tail, `succ` adds one for the head.
`average`	`⟨sum, length⟩ div`	`div(m, n) = m/n`, with `div(0, 0) = 0` so `average` is total. Traverses the list twice.
banana-split law	`⟨⦇h⦈, ⦇k⦈⟩ = ⦇⟨F(outl) h, F(outr) k⟩⦈`	Any fork of folds is a single fold, hence one traversal — `F` the base functor.
what it reduces to	`α ⟨⦇h⦈, ⦇k⦈⟩ = F(⟨⦇h⦈, ⦇k⦈⟩) ⟨F(outl) h, F(outr) k⟩`	All that (15.1a) leaves to check: the fork satisfies the defining equation, and uniqueness finishes it.
the instance	`⟨sum, length⟩ = ⦇[zeros, pluss]⦈`	`zeros = ⟨zero, zero⟩` and `pluss(a, (b, n)) = (a + b, n + 1)`, so `average` runs in one pass.
`preds`	`preds : Nat -> [Nat]`, `preds n = [n, n-1, …, 1]`	B&dM Ex 3.6 (p. 57), posed and not answered there: apply Ex 3.4 to write `preds` as `⦇k⦈ outl`.

(15.5b)



$$\langle (\!|h|\!),\ (\!|k|\!) \rangle = (\!|\langle F(\mathsf{outl})\ h,\ F(\mathsf{outr})\ k\rangle|\!)$$

The other laws.

The two $\sqsubseteq$ **fusion** rows rewrite $(\!|\alpha_B|\!)\ S$ through the same α_C and S. They are the only rows here that need the allegory **locally complete** — every hom-set a complete lattice — because they come from a least-fixed-point argument; the rest, the equality fusion (15.4a) included, needs only the initial algebra and the defining equation.

(15.5c)

name	**law**	**what it says**
Lambek	$\alpha_T{}^{\circ} = \alpha_T{}^{-1}$, the initial algebra is an isomorphism	A constructor can be undone — $\alpha_T{}^{\circ}$ takes a value apart into the parts it was built from.
Eilenberg–Wright	$\Lambda((\!|\alpha_B|\!)) = (\!|\Lambda(F(\ni)\ \alpha_B)|\!)$	A relational fold is a deterministic fold of SETS: Λ pushes the nondeterminism into the power-object, where the fold is a map again.
Eilenberg–Wright	$(\!|\alpha_B|\!) = (\!|\Lambda(F(\ni)\ \alpha_B)|\!)\ \ni$	The same fact read back — fold deterministically into a set, then take a member of it.
fusion	$(\!|\alpha_C|\!) \sqsubseteq (\!|\alpha_B|\!)\ S \Longleftarrow F(S)\ \alpha_C \sqsubseteq \alpha_B\ S$	Half of fusion: an inclusion between the two algebras is inherited by the folds.
fusion	$(\!|\alpha_B|\!)\ S \sqsubseteq (\!|\alpha_C|\!) \Longleftarrow \alpha_B\ S \sqsubseteq F(S)\ \alpha_C$	The other half, with both inclusions turned around.

The two Λ rows, drawn:

(15.5d)

$$\begin{tikzcd}
F\ T \arrow[r, dashed, "F(K)"] \arrow[d, "\alpha_T"'] & F(P\ B) \arrow[r, "F(\ni)"] \arrow[d, "\Lambda(F(\ni)\ \alpha_B)"] & F\ B \arrow[d, "\alpha_B"] \\
T \arrow[r, dashed, "K"'] & P\ B \arrow[r, "\ni"'] & B
\end{tikzcd}$$

The left square is that catamorphism's own defining square, $K = (\!|\Lambda(F(\ni)\ \alpha_B)|\!)$, and the right one is Λ's cancellation, so the outer rectangle says $K\ \ni$ satisfies the defining equation of $(\!|\alpha_B|\!)$ — and uniqueness finishes it.

§15.5.1. Fokkinga's mutual recursion theorem

- `F-Alg(𝓐)` — the algebras and their homomorphisms — has binary products whenever 𝓐 does, and the forgetful `U : F-Alg(𝓐) → 𝓐` **creates** them: the product of `h : F A → A` and `k : F B → B` is carried by `A × B`, with structure `⟨F(outl) h, F(outr) k⟩ : F(A × B) → A × B`.
- `outl` and `outr` are homomorphisms out of it, and `⟨p, q⟩ : X → A × B` is a homomorphism **iff** `p` and `q` both are — substitute the structure and compare the two forks component by component.
- The banana-split law of (15.5a) IS that product's universal property read at the initial algebra: `⦇h⦈` and `⦇k⦈` are the unique homomorphisms to `(A, h)` and `(B, k)`, so their fork is the unique homomorphism into the product, which is `⦇⟨F(outl) h, F(outr) k⟩⦈`.
- Fokkinga's theorem is **strictly more general** and is not that product: in (15.5.1a) the two arrows leave `F(A × B)`, not `F A` and `F B`, so each sees BOTH components. The product is the case where they factor as `F(outl) h` and `F(outr) k`; B&dM's Ex 3.4 is the other special case (p. 58). What it says: **any algebra on a product carrier is folded by a fork**.

$$
\begin{tikzcd}
F\,T \arrow[r, dashed, "F(\langle f,\ g\rangle)"] \arrow[d, "\alpha"'] & F(A \times B) \arrow[d, "h"] \\
T \arrow[r, dashed, "f"'] & A
\end{tikzcd}
\qquad
\begin{tikzcd}
F\,T \arrow[r, dashed, "F(\langle f,\ g\rangle)"] \arrow[d, "\alpha"'] & F(A \times B) \arrow[d, "k"] \\
T \arrow[r, dashed, "g"'] & B
\end{tikzcd}
$$

$$\alpha\ f = F(\langle f,\ g\rangle)\ h \ \wedge\ \alpha\ g = F(\langle f,\ g\rangle)\ k \ \equiv\ \langle f,\ g\rangle = (\!|\langle h,\ k\rangle|\!) \tag{15.5.1a}$$

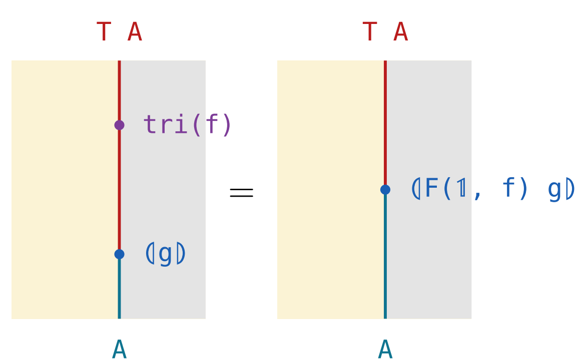
§15.5.2. Ruby triangles

(15.5.2a)

stage	definition
informally (p. 58)	$\mathtt{tri(f)}\ [a_0,\ a_1,\ \ldots,\ a_i,\ \ldots,\ a_n] = [a_0,\ f\ a_1,\ \ldots,\ f^i\ a_i,\ \ldots,\ f^n\ a_n]$
for cons-lists (p. 59)	`tri(f) = ⦇[nil, (𝟙 × listr(f)) cons]⦈`
with the base functor named	`F(A, B) = 1 + A × B`, `α = [nil, cons]`, so `tri(f) = ⦇F(𝟙, listr(f)) α⦈`
abstractly (p. 59)	`F` a bifunctor with initial type `(α, T)`: `tri(f) = ⦇F(𝟙, T(f)) α⦈`

For the definition to make sense `f : A → A` is required, and then `tri(f) : T A → T A`.

$$
\begin{tikzcd}
T\,A \arrow[rr, "\mathtt{tri(f)}"] \arrow[dr, dashed, "(\!|F(𝟙,\ f)\ g|\!)"'] & & T\,A \arrow[dl, dashed, "(\!|g|\!)"] \\
& A &
\end{tikzcd}
$$



$$\mathtt{tri(f)}\ (\!|g|\!) = (\!|F(𝟙,\ f)\ g|\!) \qquad \Longleftarrow \qquad g\ f = F(f,\ f)\ g \tag{15.5.2b}$$

(15.5.2b) is Horner's rule, generalised from cons-lists to any initial type; B&dM call it that because for cons-lists it is the schoolbook method for evaluating a polynomial (p. 58). Fusion reduces it to `F(𝟙, T(f)) α ⦇g⦈ = F(𝟙, ⦇g⦈) F(𝟙, f) g` (p. 59).

§15.5.3. Depth of a tree

(15.5.3a)

the datatype	`tree A ::= tip A | bin (tree A, tree A)`, base functor `F(A, B) = A + B × B`, initial type `([tip, bin], tree)`
the fold	`⦇[g, h]⦈` is the unique `f` with `f (tip a) = g a` and `f (bin (x, y)) = h (f x, f y)`
`tree(f)`	`tree(f) = ⦇F(f, 𝟙) [tip, bin]⦈`; pointwise `tree(f) (tip a) = tip (f a)` and `tree(f) (bin (x, y)) = bin (tree(f) x, tree(f) y)`
`max`	`max = ⦇[𝟙, bmax]⦈`, where `bmax (a, b)` is the larger of `a` and `b`
`depths`	`depths = tree(zero) tri(succ)` — replaces every tip by its depth in the tree, `zero` the constant function returning 0 and `succ` the successor
`depth`	`depth = depths max`

§16. Type functor

(16a)

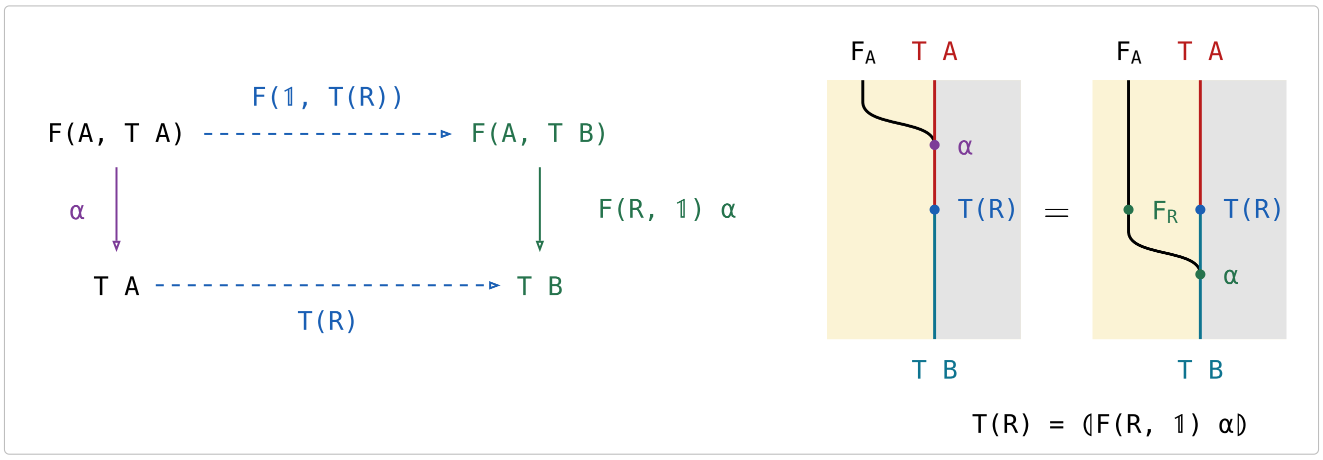
F a **binary** relator: $F(R,\ S)$ is its action on a pair, and $F(X)$ abbreviates $F(\mathbb{1},\ X)$, the F of the catamorphism section. For every object A the initial algebra is $\alpha\ :\ F(A,\ T\ A) \to T\ A$, among the maps. The **type functor** T acts on an arrow $R\ :\ A \to B$ by

$$T(R)\ =\ \llparenthesis F(R,\ \mathbb{1})\ \alpha \rrparenthesis\ :\ T\ A \to T\ B$$

F is the **base functor** — one layer of the structure, acting on the recursive position — while T is the datatype itself, acting on the parameter. For cons-lists, $\mathrm{list}(R)\ =\ \llparenthesis[\mathrm{nil},\ (R \otimes \mathbb{1})\ \mathrm{cons}]\rrparenthesis$, which is $\mathrm{map}(R)$.

A bifunctor puts its two arguments in two different places. The wire is the partial application $F_A\ =\ F(A,\ -)$, and an arrow in the SECOND argument is a bead on the object wire with the F wire running past it — the catamorphism section's rule, unchanged — while an arrow $R\ :\ A \to B$ in the FIRST induces a natural transformation $F_R\ :\ F_A \Rightarrow F_B$, which is a bead ON the F wire. Marsden draws bifunctors exactly so (`CatString.pdf`, **Bifunctors**): side-by-side is already spent on functor composition, so one argument has to go into the wire's name.

(16b)



$T(R)$ is the unique arrow making it commute; there is no $\sqsubseteq$ in it.

Two beads at one height on two different wires are one action, here $F(R,\ T(R))$, so the right panel reads $F(R,\ T(R))\ \alpha$ and the left one $\alpha\ T(R)$. Their relative height says nothing: slide F_R below the $T(R)$ bead and the panel reads $F(\mathbb{1},\ T(R))\ F(R,\ \mathbb{1})$ and then α, which is the square's top row followed by its right vertical; slide it above and the panel reads $F(R,\ \mathbb{1})\ F(\mathbb{1},\ T(R))$, the route through $F(B,\ T\ A)$ the square leaves out. Identifying those two is the exchange condition every bifunctor satisfies (Hinze & Marsden, Ex 1.23), and in this calculus there is nothing to identify.

(16c)

name	law	what it says
the defining equation	$T(R)\ =\ \llparenthesis F(R,\ \mathbb{1})\ \alpha \rrparenthesis$	Rebuild the structure with α, applying R to the parameter on the way; that is $\mathrm{map}(R)$.
functor	$T(\mathbb{1})\ =\ \mathbb{1}$ and $T(R)\ T(S)\ =\ T(R\ S)$	Mapping the identity changes nothing, and two maps in a row are one map.
type functor fusion	$T(R)\ \llparenthesis Q \rrparenthesis\ =\ \llparenthesis F(R,\ \mathbb{1})\ Q \rrparenthesis$	A map followed by a fold is a single fold — the mapped structure is never built.
naturality of α	$\alpha\ T(R)\ =\ F(R,\ T(R))\ \alpha$	Building and then mapping is the same as mapping the parts and then building, so α is natural from $G(R)\ =\ F(R,\ T(R))$ to T.
type relator	$T(R)^\circ\ =\ T(R^\circ)$	A datatype acts on relations, not only on maps — the map of the converse is the converse of the map.

Type functor fusion is the equality fusion (15.4a) applied to $T(R)$'s own defining algebra, whose side condition holds because F is a bifunctor — $F(R,\ \mathbb{1})\ F(\mathbb{1},\ \llparenthesis Q \rrparenthesis)\ =\ F(R,\ \llparenthesis Q \rrparenthesis)\ =\ F(\mathbb{1},\ \llparenthesis Q \rrparenthesis)\ F(R,\ \mathbb{1})$ — so it too needs only the initial algebra and the defining equation.

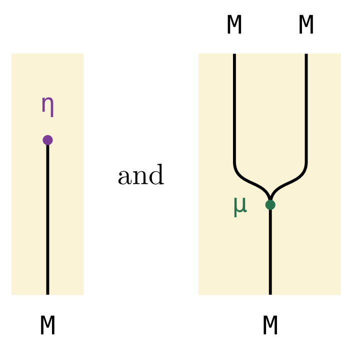
$$\begin{tikzcd}
\mathtt{F(A,\ T\ A)} \arrow[r, "\mathtt{F(R,\ T(R))}"] \arrow[d, "\alpha"'] & \mathtt{F(B,\ T\ B)} \arrow[d, "\alpha"] \\
\mathtt{T\ A} \arrow[r, "\mathtt{T(R)}"'] & \mathtt{T\ B}
\end{tikzcd}$$ (16d)

It commutes strictly, and it is (16b) with the right vertical's two halves separated: push `F(R, 𝟙)` up into the top row, where it meets `F(𝟙, T(R))` as the one action `F(R, T(R))`, and `α` is left as the vertical. In the string diagram that is the $\mathsf{F_R}$ bead sliding up past `T(R)`, and what stays put is `α`, one bead falling past `T(R)` from the row above to the row below — which is what "`α` is natural" means.

§17. Monad

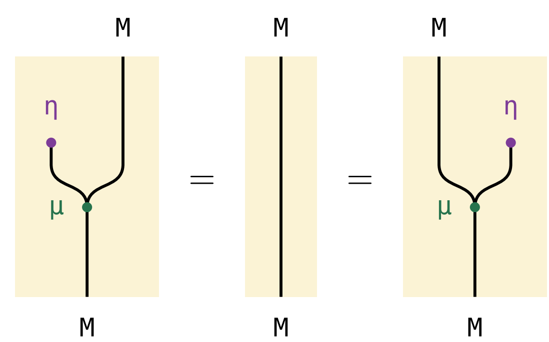
An endofunctor `M : 𝓐 → 𝓐` with two natural transformations, the **unit** `η : Id ⇒ M` and the **multiplication** `μ : M∘M ⇒ M`, subject to a unit law and an associativity law: (17a)

`(η∘M) μ = 𝟙 = (M∘η) μ` `(μ∘M) μ = (M∘μ) μ`



(17b)

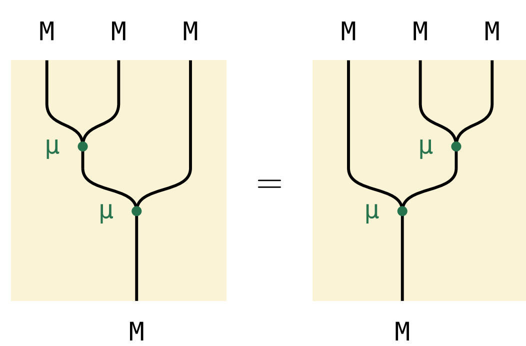
In (17b) the identity functor is a REGION and not a wire, so `η : Id ⇒ M` has nothing above it and the `M` wire simply begins at its bead — the book's lollipop. `μ : M∘M ⇒ M` is the two `M` wires of `M∘M` merging into one, its tuning fork.



(17c)

`(η∘M) μ = 𝟙 = (M∘η) μ`

The lollipop is absorbed into the fork from either side and a bare wire is left, which is what `𝟙` is in this calculus: nothing is drawn for it.



(17d)

`(μ∘M) μ = (M∘μ) μ`

The two nestings of the fork agree, so three `M`s collapse to one whichever pair is multiplied first.

§17.1. Monad in 𝒮et

(17.1a)

monad	η	μ	η on elements	μ on elements
Id A = A	𝟙	𝟙	η A a = a	μ A a = a
Exc A = A + E	ι_l	$𝟙 \triangledown \iota_r$	$\eta\ A\ a = \iota_l\ a$	$\mu\ A\ (\iota_l\ y) = y$ $\mu\ A\ (\iota_r\ e) = \iota_r\ e$
$\mathtt{State}\ A = (A \times P)^P$	curry 𝟙	$(\mathtt{apply}_{A \times P})^P$	η A a = λ x . (a, x)	μ A f = λ x . g y where (g, y) = f x
Pow A = P A	{·}	⋃	η A a = {a}	μ A Xs = ⋃ Xs

§17.2. The state monad

For a fixed set P of states, IntroString p. 67 defines

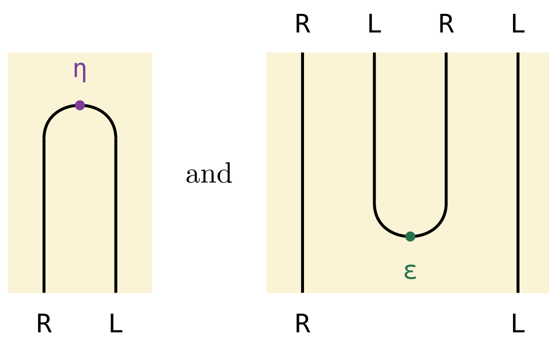
$$\mathtt{State}\ A = (A \times P)^P \qquad \eta\ A = \mathtt{curry}\ (𝟙 : A \times P \to A \times P) \qquad \mu\ A = (\mathtt{apply}_{A \times P})^P$$

Both arrows come from the curry adjunction L ⊣ R, L A = A × P and $R\ B = B^P$ (IntroString p. 96):

- State = R∘L.
- η **is** the unit Id ⇒ R∘L, which is why it is the curried identity.
- the counit ε : L∘R ⇒ Id is apply.
- μ = R∘ε∘L, that counit with a functor on each side: in $\mu\ A = (\mathtt{apply}_{A \times P})^P$ the subscript A × P is L A, the functor INSIDE, and the outer $(-)^P$ is R, the functor OUTSIDE.
- ⊣ composes nothing; it only names the left adjoint first.

Applicative order stands R left and L right in State = R∘L. The identity functor is a region, so a unit Id ⇒ R∘L has nothing above it and two wires below — one strand that OPENS the pair; a counit L∘R ⇒ Id is that strand upside down, CLOSING a pair that arrives from above.

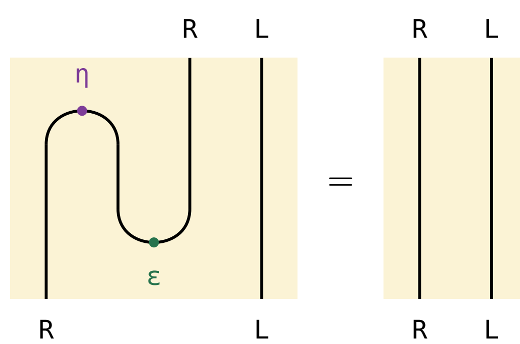
(17.2a)



In (17.2a), State∘State is the four wires R L R L, and μ = R∘ε∘L closes the middle pair — the L∘R — leaving the outer R and L, which is State again. The two wires that run past the turn untouched are the two functors composed on, R outside and L inside.

(17.2b)

$$\begin{tikzcd} \mathtt{State}\ A \arrow[r, "\eta\ (\mathtt{State}\ A)"] \arrow[dr, "𝟙"'] & \mathtt{State}\ (\mathtt{State}\ A) \arrow[d, "\mu\ A"] \\ & \mathtt{State}\ A \end{tikzcd}$$



(η∘State) μ = 𝟙

(17.2b) is the zig-zag. η∘State opens a new R L left of the pair already there and μ closes the middle two — the new L against the old R — leaving the new R and the old L, which is State unchanged. Strip the old L, which never moves, and what zig-zags is the curry adjunction's snake equation (η∘R)(R∘ε) = 𝟙 (IntroString p. 93): (17c) read at State is that triangle identity with L composed on the inside.

In Set it is beta-reduction. η (State A) f = λ x . (f, x) pairs the computation with the state, and μ A applies it back, λ x . f x, which is f.

§18. Minimum and maximum

(18a)

For `R : A → A`, `min R ≜ ∋ ∩ (∋°\R) : P A → A`, `max R ≜ min R°`.

`x (min R) a ⟺ a ∈ x ∧ (∀b ∈ x. b R a)`

`R` points from an element to the ones at most as large, so `min R` picks the element of `x` that every element of `x` points to.

(18b)

the law	B&dM
`X ⊑ min R ⟺ X ⊑ ∋` and `∋° X ⊑ R`	p. 166
`{·} (∋°\R) = R`	(7.1)
`Λ(S) (∋°\R) = S°\R`	(7.2)
`union (∋°\R) = ∋°\(∋°\R)` `union ≜ Λ(∋ ∋) : P(P A) → P A`	(7.3)
`{·} min R = 𝟙 ∩ R`	(7.4)
`Λ(S) min R = S ∩ (S°\R)`	(7.5)
`Λ(S) min R = Λ(S) min(R ∩ S° S)`	(7.6)
`E(S) min R = (∋ S) ∩ ((∋ S)°\R)`	(7.7)
`P(f) min R = min(f R f°) f`	(7.8)
`P(S) min R = (∋ S) ∩ (∋°\(S R))` `R` reflexive	(7.9)
`P(S) min R ⊑ (∋ S) ∩ (∋°\(S R))`	(7.10)
`P(min R) min R ⊑ union min R` `R` a preorder	(7.11)
`P(min R) min R = P(Dom (min R)) union min R` `R` a preorder	(7.12)

§18.1. `{·} (∋°\R) = R`

(18.1a)

$$\frac{\mathtt{X}}{\mathtt{\{\cdot\}\ (\ni^\circ\backslash R)}} \underset{\mathtt{f^\circ\cdot\dashv f\cdot}}{\Longleftrightarrow} \frac{\mathtt{\{\cdot\}^\circ\ X}}{\mathtt{\ni^\circ\backslash R}} \underset{\mathtt{T\cdot\dashv T\backslash}}{\Longleftrightarrow} \frac{\mathtt{\ni^\circ\ \{\cdot\}^\circ\ X}}{\mathtt{R}} \underset{\circ}{\Leftrightarrow} \frac{\mathtt{(\{\cdot\}\ \ni)^\circ\ X}}{\mathtt{R}} \underset{\mathtt{i\dashv E}}{\Longleftrightarrow} \frac{\mathtt{X}}{\mathtt{R}}$$

§18.2. `Λ(S) (∋°\R) = S°\R`

(18.2a)

$$\frac{\mathtt{X}}{\mathtt{\Lambda(S)\ (\ni^\circ\backslash R)}} \underset{\mathtt{f^\circ\cdot\dashv f\cdot}}{\Longleftrightarrow} \frac{\mathtt{\Lambda(S)^\circ\ X}}{\mathtt{\ni^\circ\backslash R}} \underset{\mathtt{T\cdot\dashv T\backslash}}{\Longleftrightarrow} \frac{\mathtt{\ni^\circ\ \Lambda(S)^\circ\ X}}{\mathtt{R}} \underset{\circ}{\Leftrightarrow} \frac{\mathtt{(\Lambda(S)\ \ni)^\circ\ X}}{\mathtt{R}} \underset{\mathtt{\cdot\ni\dashv\Lambda}}{\Longleftrightarrow} \frac{\mathtt{S^\circ\ X}}{\mathtt{R}} \underset{\mathtt{T\cdot\dashv T\backslash}}{\Longleftrightarrow} \frac{\mathtt{X}}{\mathtt{S^\circ\backslash R}}$$